

MATH 14–Skeletal Notes

MATH 14–Skeletal Notes

Mehdi Ahmadi

October 1, 2026

How to Use These Notes

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

Wherever a derivation, a solution, or a sketch belongs, there is a blank box, and in front of it a short parenthetical instruction saying what has to be done there — which formula to reach for, what to watch out for, what to conclude — but not how to do it. The figures, the animations, and the interactive applets that follow a box are the check on what you wrote in it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the “PDF” button in the navigation bar gives you the whole book as one file to print from. That file is rebuilt every time new notes are posted, so download a fresh copy rather than printing from an old one. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

The assignments, with their answer boxes and the AI tutor, and the review problem sets live online only, in the chapter *Assignments and Review Problems* of the web version of this book at mahmadi-ops.github.io/MATH14-Skeletal; they are meant to be worked there, not on paper, so they are not in the PDF.

The big picture

The map below gathers every integral of this book on one page. Parametrization feeds the line integrals; the five theorems act as bridges, trading an integral over a boundary for an integral over the region, surface, or solid it bounds; and each multiple integral carries its coordinate substitutions. An arrow between two topics means the second builds on the first, and the gold arrows mark the theorem and substitution bridges between the integral types. Every topic on the map is a link to the part of the book where it is covered. For comfortable reading, [open the map full size in a new tab](#). Interactive map of the book's integrals (view in the HTML version).

Contents

How to Use These Notes	iv
The big picture	v
1 Line Integrals	1
1.1 Parametrization of Curves	1
1.1.1 Parametric Equations	2
1.1.2 Some Examples of Parametrizing Curves	4
1.1.3 Intersection Points versus Collision Points	5
1.1.4 Curves in Space and Vector Functions	9
1.1.5 Tangent Lines to Curves in Space	13
1.2 Arc Length of a Curve	16
1.2.1 Arc Length in 2D	16
1.2.2 Arc Length in 3D	20
1.2.3 Activity: Where a Plane Slices a Cone	22
1.3 Line Integrals	26
1.3.1 Line Integrals of Scalar Functions	26
1.3.2 A Geometric Interpretation	32
1.3.3 Concept Check	36
1.4 Vector Fields and Line Integrals	40
1.4.1 Vector Fields	41
1.4.2 Work (Reminder)	43
1.4.3 Work of a Force Field	43
1.4.4 Concept Check	51
1.5 Path Independence, Conservative Fields, and Potential Functions	54
1.5.1 A Brief Reminder (Gradient Fields)	54
1.5.2 Path Independence and Some Basic Definitions	55
1.5.3 Line Integrals in Conservative Fields	56
1.5.4 Finding a Potential Function	59
1.5.5 Conservative Fields are Gradient Fields	62
1.5.6 The Loop Property of Conservative Fields	64
1.5.7 Concept Check	66

Chapter 1

Line Integrals

1.1 Parametrization of Curves

Objectives

- Describe the motion of a point in the plane with a pair of parametric equations $x = f(t)$ and $y = g(t)$, and understand the parameter t as recording *when* the point visits each place on the curve.
- Sketch a parametric curve from its parametric equations, and indicate the direction of increasing t in which the curve is traced.
- Find parametric equations for common curves, such as lines, circles, and ellipses, and eliminate the parameter to recover a Cartesian equation when possible.
- Distinguish the **intersection points** of two parametric curves from their **collision points**, where two moving objects reach the same place at the same time.
- Write a space curve as a vector function $\mathbf{r}(t)$, and parametrize the curve in which two surfaces meet.
- Differentiate a vector function to get the tangent vector of a space curve, and write the equation of the tangent line at a point of the curve.

Imagine that you would like to describe the motion of an object whose motion is confined to a plane. If you know how its Cartesian coordinates, x and y , change as time passes, you will be able to draw a curve that represents the trajectory of its motion. Mathematically, we will need two equations to demonstrate how the x - and y -coordinates evolve in time t , which are known as the parametric equations. This section reviews that language and then extends it to curves in space, where it becomes the main tool for everything that follows.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1.1 Parametric Equations

Definition 1.1.1 Parametric Equations. If x and y are given as functions of a third variable t , called a **parameter**, by

$$x = f(t), \quad y = g(t), \quad (1.1.1)$$

then the points $(x, y) = (f(t), g(t))$ trace out a curve in the plane, called a **parametric curve**. The equations $x = f(t)$ and $y = g(t)$ are the **parametric equations** of the curve. $\diamond$

As t runs over its interval, the point $(f(t), g(t))$ moves, and the trail it leaves behind is the curve. The parameter therefore carries more information than the picture does: it also records *when* the moving point is at each place, as in [Figure 1.1.2](#).

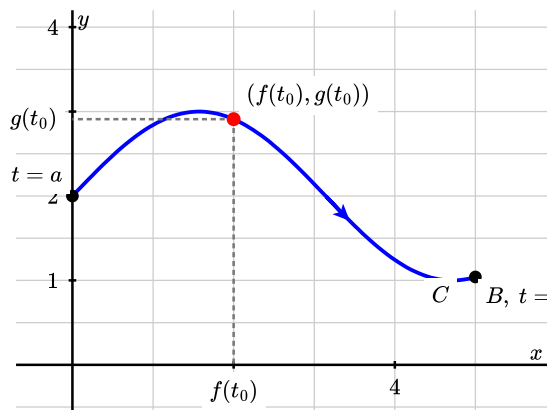
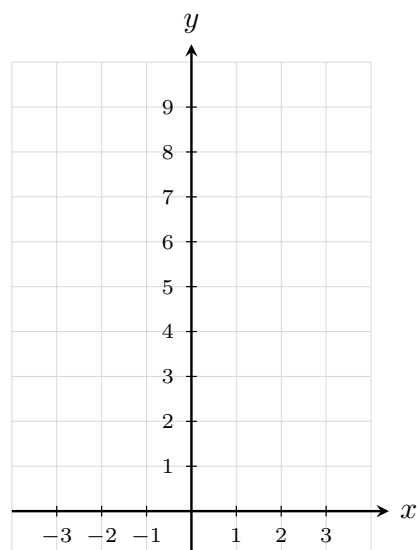


Figure 1.1.2 The trajectory of a moving point whose coordinates are given by the parametric equations $x = f(t)$, $y = g(t)$. At the instant t_0 the point is at $(f(t_0), g(t_0))$.

As an example, consider $x = t$ and $y = t^2$. Fill in [Table 1.1.3](#), plot the seven points on the grid below, and join them. (Then eliminate the parameter to name the curve, and mark on your sketch the direction in which it is traced as t increases.)

Table 1.1.3 Points on the curve $x = t$, $y = t^2$

t	x	y
-3		
-2		
-1		
0		
1		
2		
3		



Eliminating t :

Name of the curve, and its direction:

As our second example, consider the equation of the circle centered at $(0, 0)$ with unit radius,

$$x^2 + y^2 = 1. \quad (1.1.2)$$

Here, we would like to parametrize x and y in terms of a single parameter t , as in (1.1.1), in such a way that by varying the value of t we can reconstruct the circle (1.1.2). Find such a pair.

the parametrization:

interval of t , and the check:

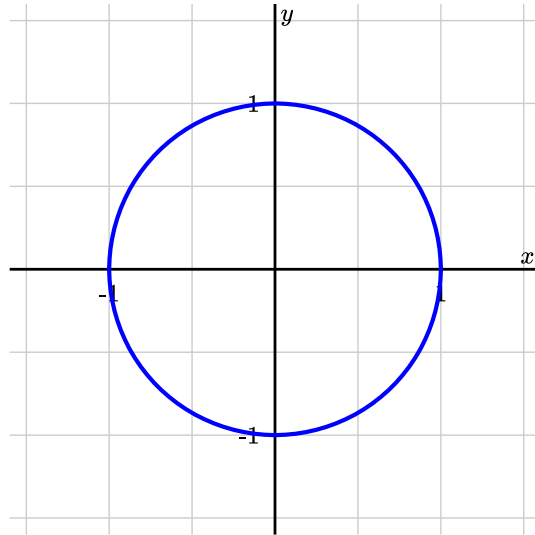


Figure 1.1.4 The unit circle $x^2 + y^2 = 1$.

Checkpoint 1.1.5 Question. Can you think of another way to write a parametric form of the circle $x^2 + y^2 = 1$?

workspace

1.1.2 Some Examples of Parametrizing Curves

Example 1.1.6 Parametrizing Curves. Find the parametric equations corresponding to the following curves.

1. The line segment connecting the two points $(1, 0)$ and $(0, 1)$.
2. The ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.
3. The circle of radius 2 centered at $(2, 0)$.

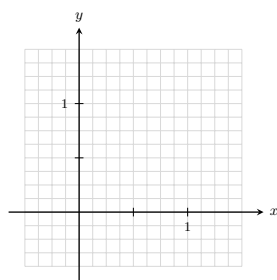
Solution.

A.

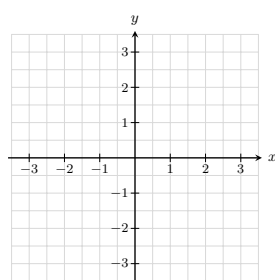
B.

C.

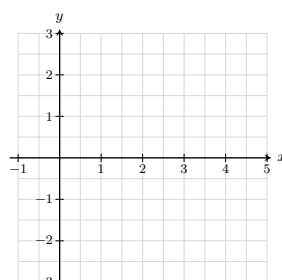
Now sketch the three curves on the grids below, marking the point reached at $t = 0$ and the direction of increasing t on each.



A.



B.



C.

□

1.1.3 Intersection Points versus Collision Points

A parametrization carries more information than the curve it traces: it also records *when* the moving point visits each place on the curve. Because of this, there are two different questions we can ask about two parametric curves

$$\mathbf{r}_1(t) = \langle f_1(t), g_1(t) \rangle,$$

$$\mathbf{r}_2(t) = \langle f_2(t), g_2(t) \rangle,$$

thought of as the trajectories of two particles.

Definition 1.1.7 Intersection Points and Collision Points. A point P is an **intersection point** of the two curves if P lies on both curves; that is, if there are parameter values t and s , not necessarily equal, with

$$\mathbf{r}_1(t) = \overrightarrow{OP} = \mathbf{r}_2(s).$$

A point P is a **collision point** of the two particles if both particles are at P at the same time; that is, if there is a single value of t with

$$\mathbf{r}_1(t) = \mathbf{r}_2(t) = \overrightarrow{OP}.$$

◇

An intersection point is a statement about the two *paths*: they cross, like two roads on a map. A collision point is a statement about the two *motions*: the cars are at the crossroads at the same instant.

Checkpoint 1.1.8 True or False. Decide whether each statement is true or false.

- (a) Every collision point of two particles is an intersection point of their paths.
- (b) Every intersection point of two paths is a collision point of the particles that travel along them.

(a) true or false, and why:

(b) true or false, and why:

Checkpoint 1.1.9 True or False. In practice the distinction dictates how we set up the equations. Decide whether each statement is true or false.

- (a) To find the **intersection points** of the curves $\mathbf{r}_1(t) = \langle f_1(t), g_1(t) \rangle$ and $\mathbf{r}_2(t) = \langle f_2(t), g_2(t) \rangle$, give the two curves different parameter names and solve $f_1(t) = f_2(s)$, $g_1(t) = g_2(s)$ for the pair (t, s) .
- (b) To find the **collision points** of the two particles, eliminate the parameters and intersect the two Cartesian curves.

(a) true or false, and why:

(b) true or false, and why:

Example 1.1.10 Two particles move in the plane with position vectors

$$\mathbf{r}_1(t) = \langle t, t^2 \rangle,$$

$$\mathbf{r}_2(t) = \langle -t, 2 - t \rangle, \quad t \in \mathbb{R}.$$

Find all intersection points of their paths, and all collision points of the particles.

Solution.

the two paths and where they meet:

collision equations:

when each particle reaches each crossing:

□

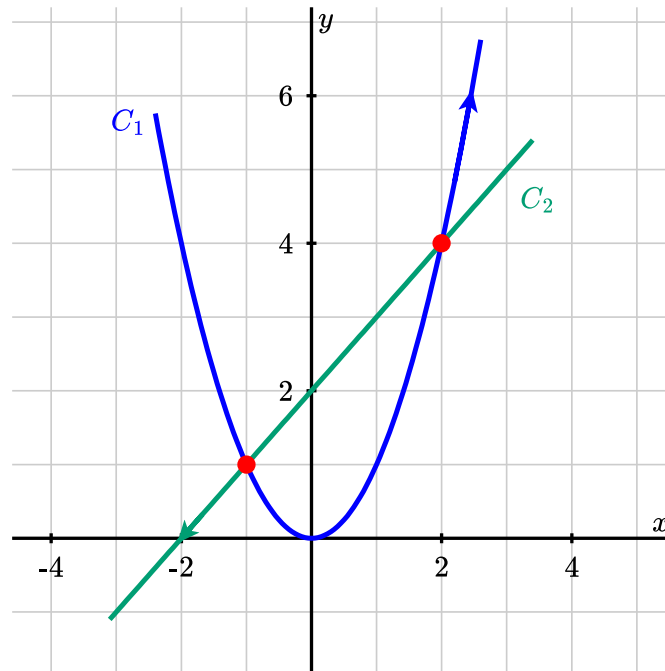


Figure 1.1.11 The parabola C_1 and the line C_2 ; compare with what you found.

Example 1.1.12 Keep the first particle, but re-schedule the second one:

$$\begin{aligned} \mathbf{r}_1(t) &= \langle t, t^2 \rangle, \\ \mathbf{r}_2(t) &= \langle 2t - 2, 2t \rangle, \quad t \in \mathbb{R}. \end{aligned}$$

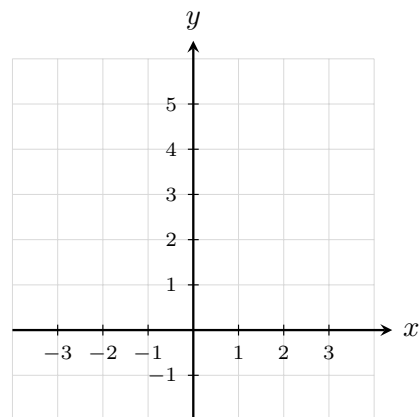
Find the intersection points of the paths and the collision points of the particles.

Solution.

the paths:

collision equations:

the other crossing:



□

Checkpoint 1.1.13 Keep $\mathbf{r}_1(t) = \langle t, t^2 \rangle$. Find a parametrization $\mathbf{r}_2$ of the line $y = x + 2$ for which the two particles collide at *both* intersection points.

$\mathbf{r}_2(t) =$

the collision equations it produces:

1.1.4 Curves in Space and Vector Functions

Assume that we are interested in describing the path of an object in space. To this aim, we can specify its location at each instant of time; that is, we can write its coordinates in $\mathbb{R}^3$ as

$$x = f(t), \quad y = g(t), \quad z = h(t). \quad (1.1.3)$$

Here t is the time parameter, which varies from some initial time t_i to some final time t_f , that is, $t \in [t_i, t_f]$. Nothing has changed, except that where a plane curve needed the two equations (1.1.1), a space curve needs three.

There is a second, more compact way to look at (1.1.3), and it is the point of view we will use for the rest of the course. Instead of three separate coordinate equations, collect the coordinates into a single **vector function** — a function whose input is a real number and whose output is a vector:

$$\begin{aligned} \mathbf{r}(t) &= \langle f(t), g(t), h(t) \rangle \\ &= f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}, \quad t \in [t_i, t_f]. \end{aligned} \quad (1.1.4)$$

Think of $\mathbf{r}(t)$ as the **position vector** of the moving object: an arrow drawn from the origin to the object's location at time t . As t increases, the arrow swings around and changes length, and its *tip* sweeps out the curve. So a parametrization can be read in two equivalent ways.

- **Coordinatewise.** Three ordinary scalar functions f, g, h of one variable, one for each coordinate, as in (1.1.3).
- **Vectorially.** One vector-valued function $\mathbf{r}$ of one variable, as in (1.1.4). The curve is the set of tips of the vectors $\mathbf{r}(t)$.

The two descriptions carry exactly the same information, but the vector form is the more useful one: it lets us differentiate and integrate a whole motion at once, and its derivative will turn out to be the velocity of the moving object. The same device works in the plane, where $\mathbf{r}(t) = \langle f(t), g(t) \rangle$; the position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ of Example 1.1.10 were already written this way.

Example 1.1.14 A helix. Consider the object whose coordinates are parametrized by

$$x = \cos(2t), \quad y = \sin(2t),$$

$$z = t, \quad t \in [0, \pi]. \quad (1.1.5)$$

Write the motion as a vector function and describe the path.

Solution.

$\mathbf{r}(t) =$

the shadow on the xy -plane, and how often it goes around:

what z does:

the path, its start, and its end:

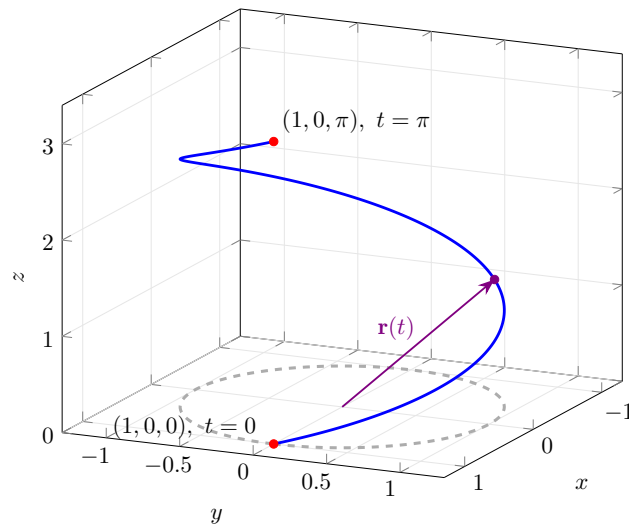


Figure 1.1.15 One turn of the helix $\mathbf{r}(t) = \langle \cos(2t), \sin(2t), t \rangle$, $0 \leq t \leq \pi$. The violet arrow is the position vector $\mathbf{r}(t)$ at $t = 3\pi/8$; its tip rides along the curve as t increases. The dashed gray circle is the shadow of the path on the plane $z = 0$.

The same helix, animated, and continued past $t = \pi$ to three full turns, is [Figure 1.1.16](#). Press “Play” to watch the point climb the curve, with its shadow running around the unit circle below and the position vector $\mathbf{r}(t)$ following it; or drag the slider to place the point yourself.

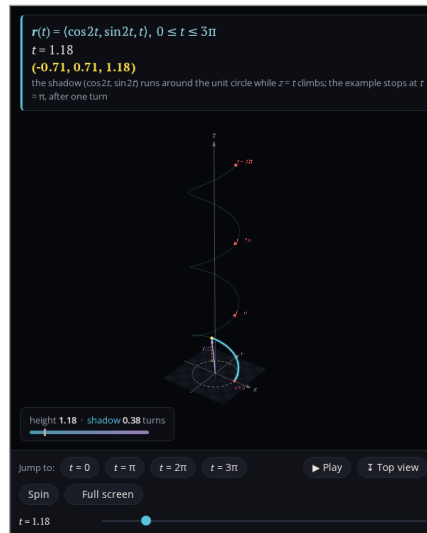


Figure 1.1.16 The helix $\mathbf{r}(t) = \langle \cos 2t, \sin 2t, t \rangle$, animated, for $0 \leq t \leq 3\pi$: the curve of the example is the first turn, up to $t = \pi$. “Top view” looks straight down the z -axis, where the whole helix collapses onto its projection, the unit circle in the xy -plane.

□

There are situations in which a curve in space arises as the intersection of two surfaces. In the next two examples we turn such a description into a vector function $\mathbf{r}(t)$.

Example 1.1.17 Where a plane cuts a cylinder. Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

Solution.

$x(t)$ and $y(t)$:

$z(t)$:

$\mathbf{r}(t) =$

check at $t = 0, \pi/2, \pi, 3\pi/2$:

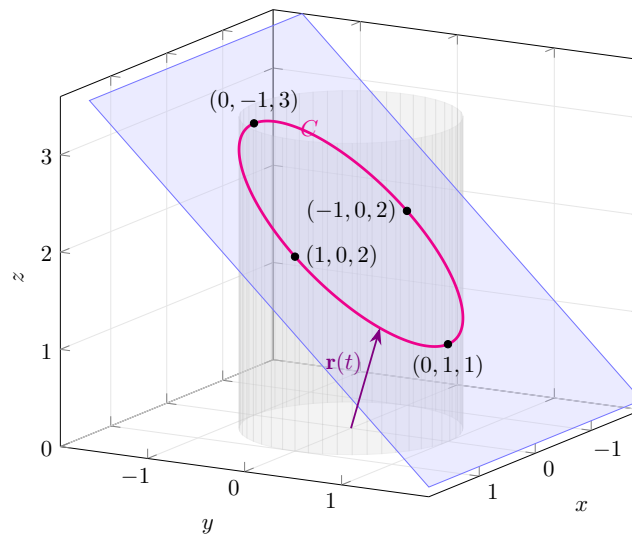


Figure 1.1.18 The cylinder $x^2 + y^2 = 1$, the plane $y + z = 2$, and the ellipse C in which they meet. The violet arrow is the position vector $\mathbf{r}(t)$ of your parametrization at $t = \pi/4$.

□

Example 1.1.19 Where two curved surfaces meet. Parametrize the intersection of the surfaces

$$z = \sqrt{1 + x^2 + y^2} \quad \text{and} \quad z = \sqrt{4 - 4x^2 - y^2}.$$

Solution.

$x(t)$ and $y(t)$:

$z(t)$, and $\mathbf{r}(t) =$

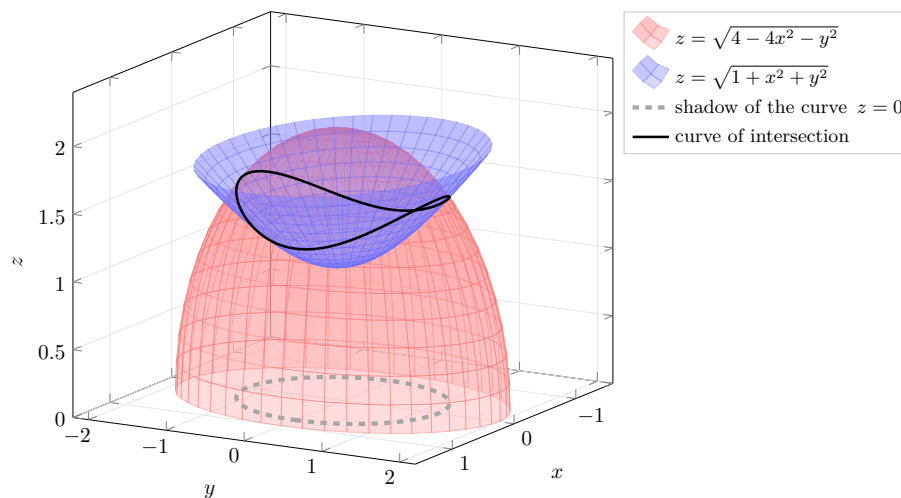


Figure 1.1.20 The upper sheet of $z = \sqrt{1 + x^2 + y^2}$ (blue) and the dome $z = \sqrt{4 - 4x^2 - y^2}$ (red) meet along the closed curve drawn in black. Its shadow in the plane $z = 0$ is the dashed ellipse; compare it with the equation you found.

□

1.1.5 Tangent Lines to Curves in Space

Now that a curve in space is a vector function $\mathbf{r}(t)$, we can ask the same question we asked about graphs in single-variable calculus: what is the tangent line at a point of the curve? The applet in [Figure 1.1.21](#) shows how to find its direction. Fix a time t_0 , mark the point P that the moving object occupies at that time, and let h be a small change in time.

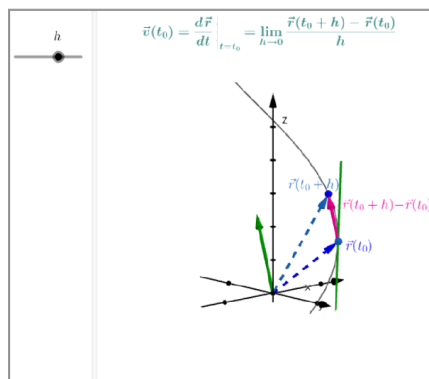


Figure 1.1.21 The tangent vector as the limit of a difference quotient. The gray curve is the helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$, and the two blue arrows are the position vectors $\mathbf{r}(t_0)$ and $\mathbf{r}(t_0 + h)$, with $t_0 = \pi/4$. The pink arrow is their difference $\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)$, the green arrow is the difference quotient $(\mathbf{r}(t_0 + h) - \mathbf{r}(t_0))/h$, and the green line is the tangent line at t_0 .

The pink arrow $\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)$ joins the point of the curve at time t_0 to the point of the curve at time $t_0 + h$: it is a *chord* of the curve. Dividing it by h gives the green arrow

$$\frac{\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)}{h}.$$

(Explain what dividing by h does to the chord, including when $h < 0$, and

why the green arrow becomes tangent to the curve at P as $h \rightarrow 0$; compare with [Figure 1.1.21.](#))

dividing the chord by h :

as $h \rightarrow 0$:

The limit of the difference quotient is therefore a vector tangent to the curve at P . We call it the **tangent vector**, or the **velocity**, of the path:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}. \quad (1.1.6)$$

The limit in [\(1.1.6\)](#) is taken one coordinate at a time, so if $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, then $\mathbf{v}(t) = \langle x'(t), y'(t), z'(t) \rangle$: to differentiate a vector function, differentiate each of its coordinates. The tangent vector at the time t_0 is

$$\mathbf{v}(t_0) = \left. \frac{d\mathbf{r}}{dt} \right|_{t=t_0}.$$

The tangent line at P is the line through P in the direction of $\mathbf{v}(t_0)$. (Write the position vector $\mathbf{r}_0$ of $P(x(t_0), y(t_0), z(t_0))$, then use the vector equation of a line from MATH 13. Use a new parameter s for the line, and say why it must differ from t .)

$\mathbf{r}_0 =$

the tangent line:

why s and not t :

The tangent line.

The tangent line to the curve $\mathbf{r}(t)$ at $t = t_0$, with $\mathbf{r}_0 = \mathbf{r}(t_0)$, is

$$\mathbf{r}(s) = \mathbf{r}_0 + s \mathbf{v}(t_0), \quad s \in \mathbb{R}. \quad (1.1.7)$$

Example 1.1.22 Tangent line to a helix. Find the tangent line to the path $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{\pi}{4} \right)$.

Solution. (First find the time t_0 at which the path passes through the point, and check all three coordinates. Then apply (1.1.7), and give the answer in both vector and parametric form; compare with the green line in Figure 1.1.21.)

the time t_0 :

$\mathbf{v}(t) =$

$\mathbf{v}(t_0) =$

vector form, $\mathbf{r}(s) =$

parametric form:

The same construction, animated, is Figure 1.1.23. Press “Play” for a guided tour, or drag h toward 0 and watch the green difference quotient turn onto the gold tangent line.

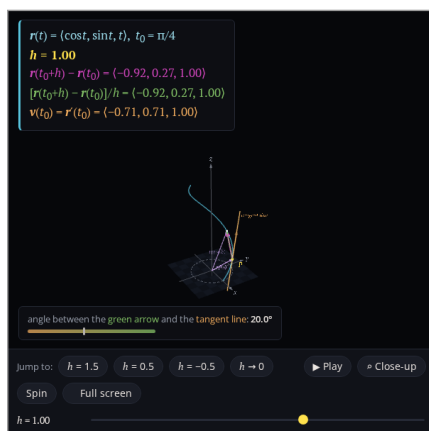


Figure 1.1.23 The tangent vector as a limit, animated. The helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ for $0 \leq t \leq 3\pi/2$, the point $P = \mathbf{r}(t_0)$ with $t_0 = \pi/4$, the position vectors $\mathbf{r}(t_0)$ and $\mathbf{r}(t_0 + h)$ (violet), the chord $\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)$ (pink), the difference quotient (green), and the tangent line at P with the tangent vector $\mathbf{v}(t_0)$ (gold).

□

1.2 Arc Length of a Curve

When an object moves along a curve, we often want to know the total distance it travels. In this section we build the arc-length integral, first in the plane and then in space, and we apply it to two motions: a point sweeping out a quarter circle and a paraglider spiralling to the ground. The key idea throughout is simple: *the total arc length of a curved path can be approximated as the sum of the arc lengths of n small pieces.*

1.2.1 Arc Length in 2D

Goal. Compute the arc length S of a curve using its parametrized form. Imagine that at time $t = a$ an object starts moving from an initial point A and at time $t = b$ it reaches the final point B . During this period the object traces out a smooth curve C in the plane. Throughout this section we illustrate the construction on the closed curve of [Figure 1.2.1](#), a circle traversed once counterclockwise, so that the object returns to where it began and A and B are the same point. How can we calculate the distance the object has travelled?

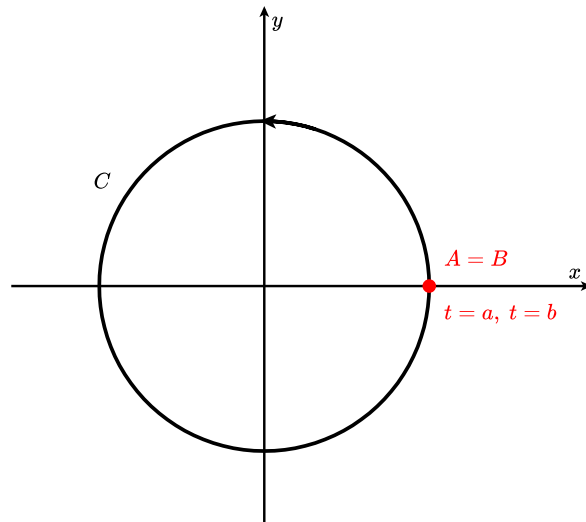


Figure 1.2.1 A smooth curve C traced by a moving object, from the starting point A at time $t = a$ to the ending point B at time $t = b$. Here C is closed, so the object returns to its starting point and $A = B$.

The strategy is to replace the curve by a chain of straight segments whose lengths we already know how to compute, as in [Figure 1.2.2](#).

workspace

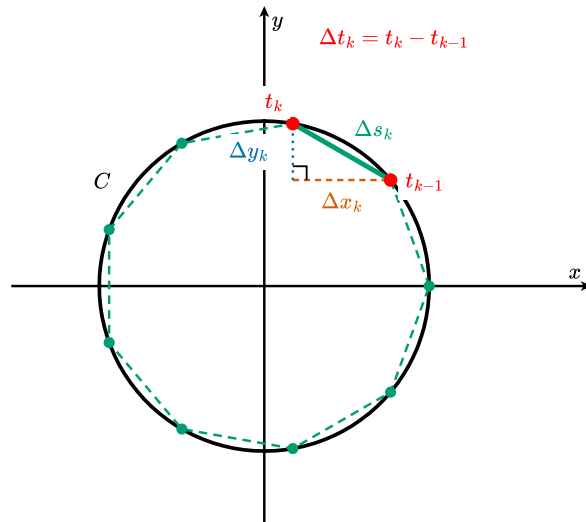


Figure 1.2.2 Approximating the length of a curve C by an inscribed polygon with n sides. The k -th chord joins the point reached at time t_{k-1} to the point reached at time t_k ; its length Δs_k is the hypotenuse of a right triangle with legs Δx_k and Δy_k . The corresponding time step is $\Delta t_k = t_k - t_{k-1}$.

Now build the arc length from the chords. (Approximate S by the sum of the chord lengths and let $n \rightarrow \infty$.)

$S \approx \text{sum of the chords:}$

$\text{the limit } n \rightarrow \infty:$

$\text{divide and multiply by } \Delta t_k, \text{ then take the limit again:}$

$\text{the square root is:}$

Results.

Let a plane curve C be parametrized by $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ for $a \leq t \leq b$. Then

$$S = \int_C ds = \int_{t=a}^{t=b} \|\mathbf{v}(t)\| dt, \quad (1.2.1)$$

where

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle, \quad ds = \|\mathbf{v}(t)\| dt. \quad (1.2.2)$$

In words: *arc length is the integral of speed over time*. This is the natural generalization of the familiar rule “distance equals speed times time” to motion whose speed and direction change from moment to moment.

Example 1.2.3 Length of a quarter circle. Using the arc-length formula (1.2.1), calculate the length of a quarter circle of radius one. (We already know the answer: a full unit circle has circumference 2π , so a quarter of it has length $\frac{\pi}{2}$. The point of this example is to see the formula reproduce it.)

Solution.

arc length swept out up to time t :

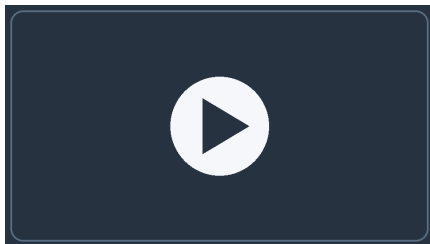


Figure 1.2.4 The point $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$ sweeps out the quarter circle as t increases from 0 to $\pi/2$; the accumulated arc length $s(t) = t$ reaches $\pi/2 \approx 1.5708$.

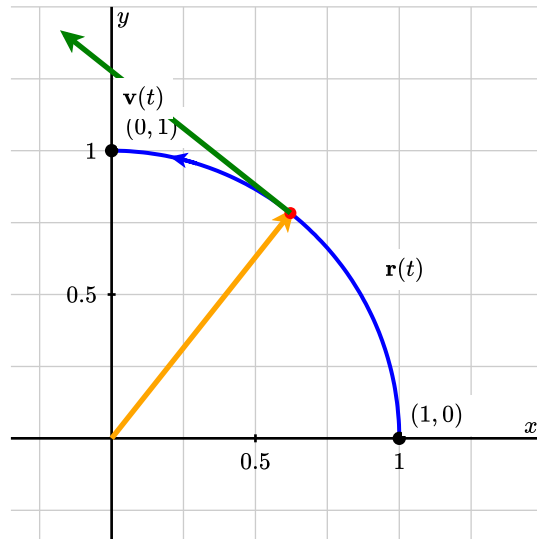


Figure 1.2.5 The quarter circle $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$ with the position vector (orange) and the unit velocity vector $\mathbf{v}(t)$ (green), which is tangent to the curve.

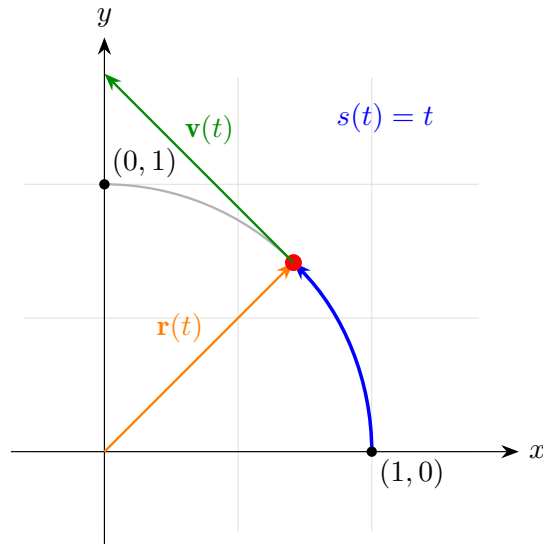


Figure 1.2.6 Snapshot at $t = \pi/4$: the arc already swept out (blue) has length $s(t) = t = \pi/4$, the part not yet traced is gray, and the unit velocity vector $\mathbf{v}(t)$ is tangent to the circle at the moving point.

□

1.2.2 Arc Length in 3D

The formulas (1.2.1) and (1.2.2) generalize to motion in space with no new ideas: the k -th chord simply picks up a third leg, so $(\Delta s_k)^2 = (\Delta x_k)^2 + (\Delta y_k)^2 + (\Delta z_k)^2$ and in the limit $ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$. For a space curve $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ we therefore have

$$S = \int_C ds = \int_{t=a}^{t=b} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \quad (1.2.3)$$

$$\begin{aligned}
&= \int_{t=a}^{t=b} \|\mathbf{v}(t)\| \, dt, \\
\mathbf{v}(t) &= \frac{d\mathbf{r}(t)}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle, \\
\|\mathbf{v}(t)\| &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}.
\end{aligned}$$

As in the plane, arc length is still the integral of speed over time.

Example 1.2.7 Distance travelled by a paraglider. The position of a paraglider at time t is

$$\mathbf{r}(t) = \left\langle t \cos t, t \sin t, 9\sqrt{8} - \frac{\sqrt{8}}{3} t^{3/2} \right\rangle.$$

Assuming that the paraglider begins at time $t = 0$, calculate the distance travelled by the paraglider until it reaches the ground, that is, $z = 0$.

Solution.

$S =$



Figure 1.2.8 The paraglider spirals down from $\langle 0, 0, 9\sqrt{8} \rangle$ to the ground. Use it to check the landing time and the distance travelled that you found.

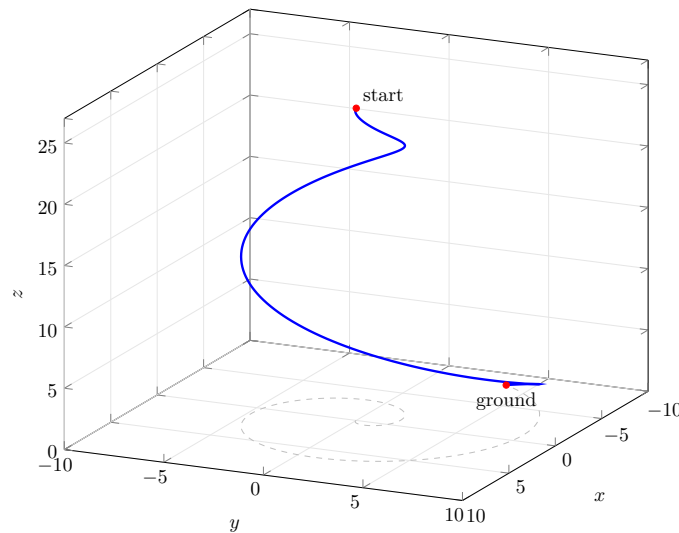


Figure 1.2.9 The paraglider's path; compare with what you found.

□

1.2.3 Activity: Where a Plane Slices a Cone

Activity 1.2.1 Where a Plane Slices a Cone. Let S_1 be the upper half of a circular cone and let S_2 be a slanted plane:

$$S_1 : z^2 = x^2 + y^2, \quad z \geq 0, \quad S_2 : z = x + 1.$$

Let $C = S_1 \cap S_2$ be their curve of intersection, shown in [Figure 1.2.10](#).

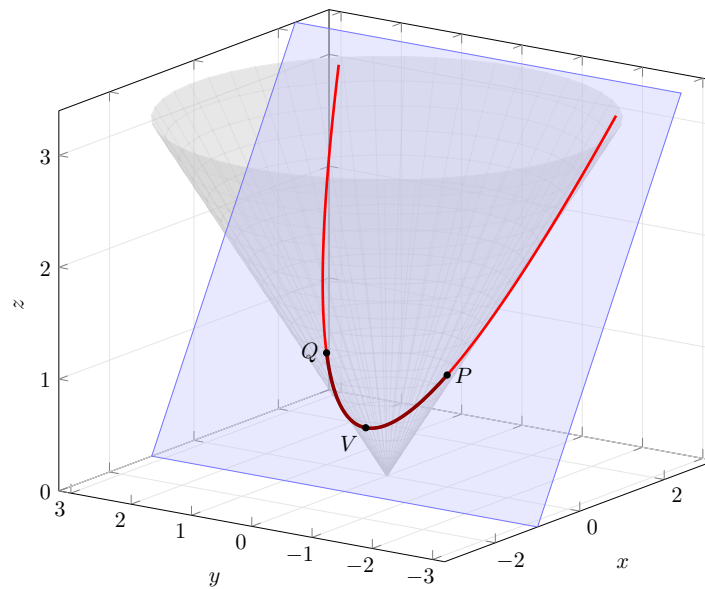


Figure 1.2.10 The cone $z^2 = x^2 + y^2$ with $z \geq 0$, the plane $z = x + 1$, and the curve C where they meet. The darker portion is the arc from $P(0, -1, 1)$ to $Q(0, 1, 1)$.

- (a) Show that every point of C satisfies $y^2 = 2x + 1$, and use this to find a vector parametrization $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ of C . Then verify directly that your $\mathbf{r}(t)$ lies on *both* surfaces for every t .

Solution.

workspace

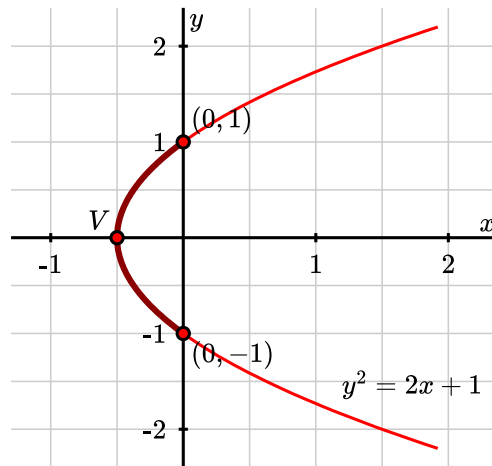


Figure 1.2.11 The shadow of C in the xy -plane is the parabola $y^2 = 2x + 1$, with vertex V . The darker arc is the projection of the piece of C that runs from P to Q .

- (b) The curve C crosses the plane $x = 0$ at the two points $P(0, -1, 1)$ and $Q(0, 1, 1)$. Set up the arc-length integral (1.2.3) for the piece of C joining P to Q and evaluate it *exactly*.

Solution. (Substitute $u = \sqrt{2}t$ to bring the integrand to the form $\sqrt{1 + u^2}$.)

t at P and at Q ; $\mathbf{v}(t) = \|\mathbf{v}(t)\| =$

$S = \int \dots$, after the substitution $u = \sqrt{2}t$:

$\int \sqrt{1+u^2} du =$ (by parts; solve for the integral)

apply the limits: $S =$

sanity check:

- (c) The point $V(-\frac{1}{2}, 0, \frac{1}{2})$ is the vertex (the lowest point) of C . Find the arc-length function $s(t)$ measured from V , and use it to compute the exact length of C from V to the point $(\frac{3}{2}, 2, \frac{5}{2})$.

Solution. (Which value of t gives V ? Start the integral there; the logarithm simplifies, because $3 + 2\sqrt{2}$ is a perfect square.)

workspace

1.3 Line Integrals

In this section we study line integrals of scalar functions. We develop the idea through a concrete problem: finding the mass of a curved wire when we know its density—its mass per unit length—at every point. The strategy is the one behind every integral: *chop the wire into tiny pieces, multiply the density of each piece by its length, and add up the results.*

1.3.1 Line Integrals of Scalar Functions

Suppose we would like to compute the mass of a wire that does not have a trivial shape (such as a straight line). If the wire were straight and its density constant, the mass would simply be density times length. For a curved wire whose density $\rho(x, y, z)$ varies from point to point, neither shortcut is available—but we can approximate mass of small portions of the wire by treating each as if it were straight and had constant density.

Assume the wire is represented by a curve C in space, parametrized by $t \in [a, b]$, as in [Figure 1.3.1](#). Divide the wire into n small portions, and denote the length of the k th portion by Δs_k . Pick a sample point (x_k, y_k, z_k) on this portion. If the portion is short, the density is nearly constant along it, so

$$\text{mass of the } k\text{th portion} \approx \text{_____} \quad (1.3.1)$$

and summing [\(1.3.1\)](#) over all portions,

$$\text{mass of the wire} \approx \text{_____} \quad (1.3.2)$$

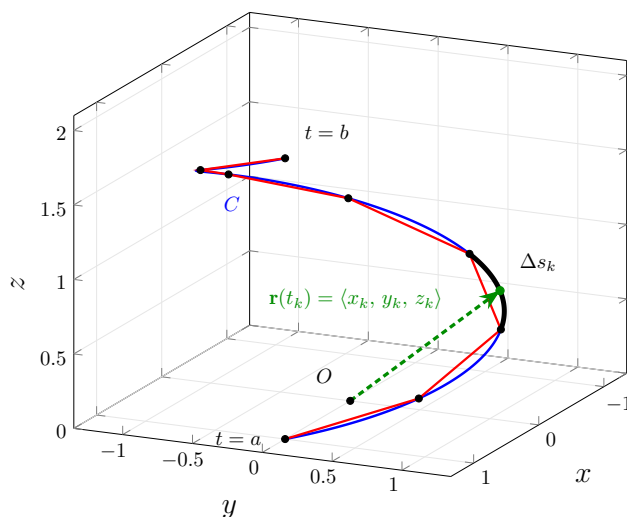


Figure 1.3.1 A wire represented by a curve C in space, parametrized by $t \in [a, b]$ and cut into $n = 7$ portions by the partition points (black dots). The k th portion (black) has length Δs_k and contains the sample point (x_k, y_k, z_k) , located by the position vector $\mathbf{r}(t_k)$. The red chords form the inscribed polygon that approximates C ; here its total length falls about 3% short of the length of C .

The estimate (1.3.2) is only an approximation, because the density is not actually constant over each portion. How can we make the calculation exact? We take the limit in which every portion becomes infinitely small, that is, $\Delta s_k \rightarrow 0$, or equivalently $n \rightarrow \infty$:

$$\text{mass of the wire} = \text{_____} \quad (1.3.3)$$

The limit in (1.3.3) is denoted by the integral

$$\text{mass of the wire} = \text{_____} \quad (1.3.4)$$

which is read as “the line integral of ρ over the curve C .” The process is animated in Figure 1.3.2: as the partition refines, the sums settle down to the value of the integral.

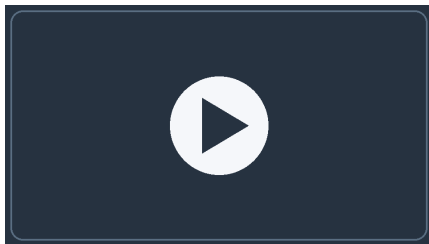


Figure 1.3.2 The construction above, on the wire C of Figure 1.3.1: the wire is cut into n portions, and the k th of them, of length Δs_k , carries a sample point (x_k, y_k, z_k) , so its mass is about $\rho(x_k, y_k, z_k) \Delta s_k$. As n grows the portions shrink, and the sum $\sum_k \rho(x_k, y_k, z_k) \Delta s_k$ settles down to the line integral $\int_C \rho \, ds$.

Our final task is to write the line element ds in terms of dt , so the integral becomes an ordinary integral we can evaluate.

$ds =$

substitute into the mass integral and factor out dt :

mass of the wire =

Results.

For a wire C parametrized by $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$, with velocity $\mathbf{v}(t) = d\mathbf{r}/dt$, the line element is

$$ds = \|\mathbf{v}(t)\| dt, \quad (1.3.5)$$

so the mass of the wire is

$$\text{mass of the wire} = \int_a^b \rho(x(t), y(t), z(t)) \|\mathbf{v}(t)\| dt. \quad (1.3.6)$$

There is nothing special about density here. In general, a **scalar line integral** has the form

$$\int_C f(x(t), y(t), z(t)) ds = \int_a^b f(x(t), y(t), z(t)) \|\mathbf{v}(t)\| dt, \quad (1.3.7)$$

where f is a scalar-valued function and the curve C specifies the region of integration. Once the curve is parametrized in terms of t , the region of integration translates to $a \leq t \leq b$. In words: *to compute a line integral, trade the geometric element ds for the parameter element $\|\mathbf{v}(t)\| dt$* . Two sanity checks connect this to what we already know. Taking $f = 1$ in (1.3.7) recovers the arc length $\int_C ds = \int_a^b \|\mathbf{v}(t)\| dt$ of (1.2.1)—total length is the “mass” of a wire of density one. And $\|\mathbf{v}(t)\|$ is exactly the speed from that section, so the factor converts time spent into distance covered.

Example 1.3.3 Mass and length of a helical wire. Consider a wire of length l and mass m . The wire is helix-shaped, with its bottom end located at

$(1, 0, 0)$ and parametric equation

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle, \quad t \geq 0. \quad (1.3.8)$$

Its density (mass per unit length) is $\rho(x, y, z) = 4x^2y^2$ kg/m. Assuming that $m = \frac{\pi}{2\sqrt{2}}$ kg, calculate the length l of the wire.

Solution.

$$\mathbf{v}(t) = \quad \text{and} \quad \|\mathbf{v}(t)\| =$$

$$\rho(x(t), y(t), z(t)) =$$

$$m = \int_0^b \cdots =$$

solve for b (and why it is the only solution):

$$l =$$

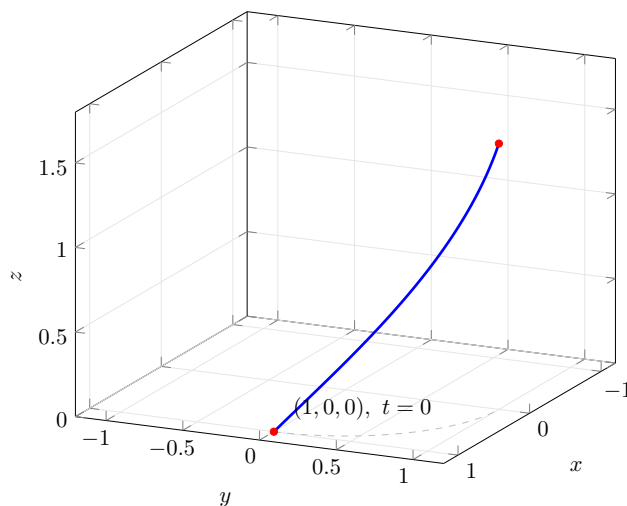


Figure 1.3.4 The helical wire $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$, from its bottom end $(1, 0, 0)$ to its top end. The dashed gray curve is its shadow on the plane $z = 0$. Compare the top end with the value of b you found.

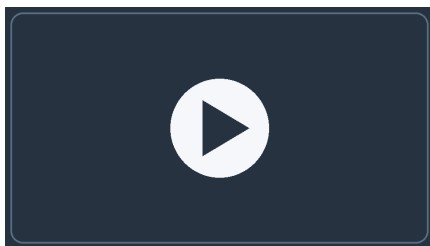


Figure 1.3.5 A point traverses the helical wire while the accumulated mass $\int_0^t \rho \|\mathbf{v}\| d\tau$ grows. Use it to check the mass, the value of b , and the length you found.

□

Example 1.3.6 A wire whose density is read off the picture. A wire of length 40 meters is parametrized as

$$\mathbf{r}(t) = \left\langle t \cos(t), t \sin(t), \frac{2\sqrt{2}}{3} t^{3/2} \right\rangle, \quad (1.3.9)$$

where $x(t)$, $y(t)$ and $z(t)$ are in meters and t is in seconds. Assume that the mass density at a point P of the wire is

$$\rho = \frac{1}{d+1} \text{ kg/m},$$

where d is the distance from P to the z -axis.

The wire is graphed in [Figure 1.3.7](#). Note that there is a piece of information that has to be extracted from the graph. Calculate the mass of the wire.

Solution.

what the graph supplies: the wire is $\leq t \leq b$ because

$$\mathbf{v}(t) = \quad \text{and} \quad \|\mathbf{v}(t)\| =$$

the upper limit: $40 =$

$$d = \quad \text{so} \quad \rho =$$

$$M =$$

what the cancellation means:

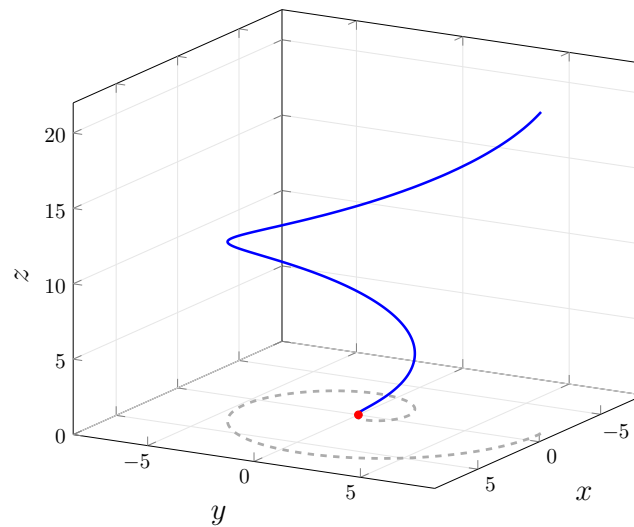


Figure 1.3.7 The wire (1.3.9), spiralling outward and upward from the origin. The dashed gray curve is its shadow on the plane $z = 0$.

□

1.3.2 A Geometric Interpretation

A line integral of a scalar function along a path that is confined to a plane has a nice geometric interpretation. We describe it first, and then work through two examples.

Let C be a curve in the xy -plane and let $f(x, y) \geq 0$ on C . The line integral $\int_C f(x, y) ds$ is equal to the surface area of the wall whose base is C and whose height above each point (x, y) of C is given by the integrand $f(x, y)$. To see this, approximate the area by dividing the wall into n slabs, as in Figure 1.3.8.

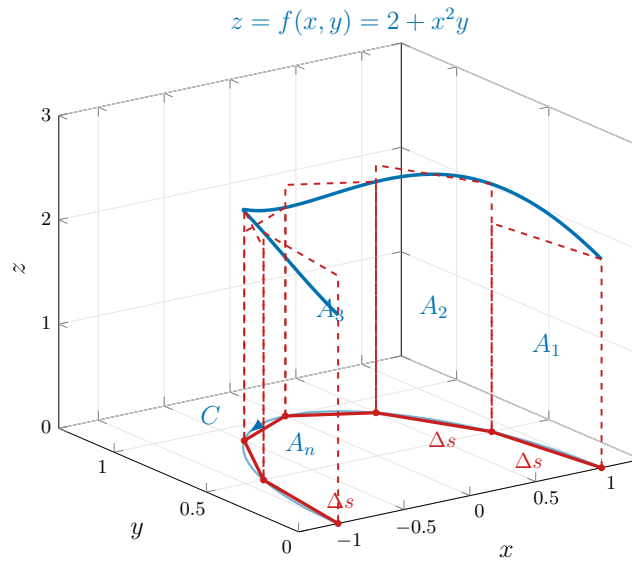


Figure 1.3.8 A wall standing on the upper half of the circle $x^2 + y^2 = 1$, with height $f(x, y) = 2 + x^2y$ (the wall of [Example 1.3.10](#)), cut into n slabs. Neighbouring points of C are joined by chords, and the k th chord, of length Δs , approximates the arc of C between its ends. The k th slab stands on this chord and is given the constant height $f(x(t_k), y(t_k))$, so its area, $f(x(t_k), y(t_k)) \Delta s$, approximates the area A_k of the wall above that arc.

area of the k th slab: $A_k \approx$

area of the wall, approximately: $A \approx$

exact area: $A =$

[Figure 1.3.9](#) builds the wall of [Example 1.3.10](#) one step at a time: the surface, then the base of the wall, then its top, and then the slabs, standing on chords of C . Drag n and watch the total area of the slabs approach the area of the wall, or press “Play” for a guided tour.

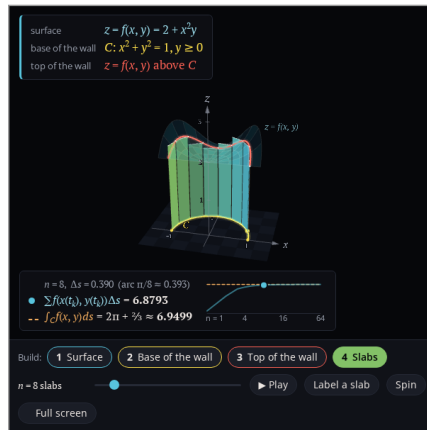


Figure 1.3.9 The wall of [Example 1.3.10](#), animated: the graph of $z = f(x, y) = 2 + x^2y$ (blue), the base of the wall, the upper half C of the circle $x^2 + y^2 = 1$ (yellow), the top of the wall, the curve on the graph directly above C (red), and n flat slabs whose total area approximates the area of the wall. Each slab stands on a chord of length Δs joining two neighbouring points of C , and has the height $f(x(t_k), y(t_k))$ of the wall at the chord's left end.

Example 1.3.10 The wall on a half circle. Calculate the line integral $\int_C (2 + x^2y) ds$, where C is the upper half of the circle $x^2 + y^2 = 1$, as shown in [Figure 1.3.11](#).

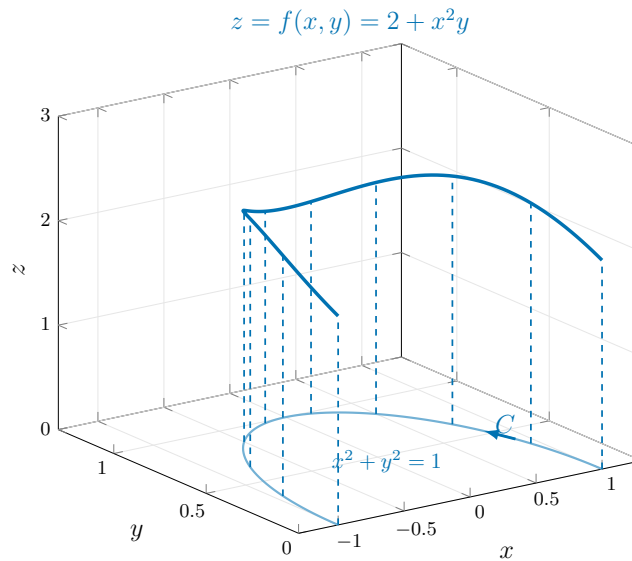


Figure 1.3.11 The curve C of [Example 1.3.10](#), the upper half of the circle $x^2 + y^2 = 1$, traversed from $(1, 0)$ to $(-1, 0)$, and the wall that stands on it: above each point of C the wall reaches the height $f(x, y) = 2 + x^2y$.

Solution.

workspace

□

Example 1.3.12 Reading the wall off the picture. Calculate the area of the wall marked with the vertical dashed blue lines in [Figure 1.3.13](#).

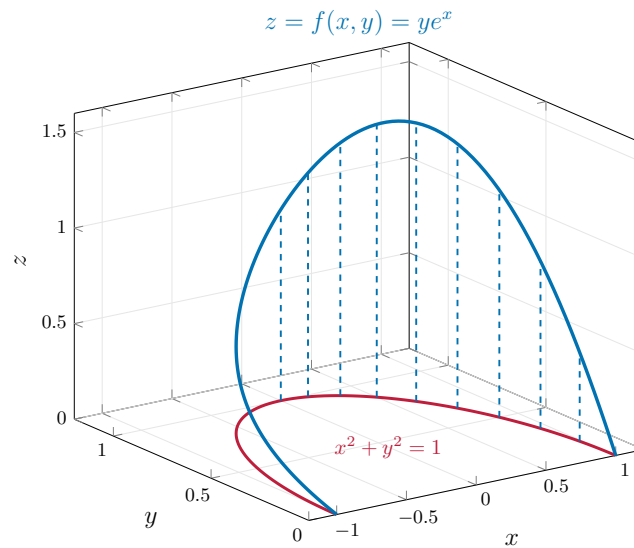


Figure 1.3.13 The graph of $z = f(x, y) = ye^x$ above the upper half of the circle $x^2 + y^2 = 1$. The wall of [Example 1.3.12](#) is the part marked by the vertical dashed lines.

Solution.

workspace

□

1.3.3 Concept Check

This last part of the section is not about computing anything new. It is about *reviewing* the concepts we have just built. Everything below refers to the two applications developed so far using line integrals, in particular the length L of a wire and its mass M ,

$$L = \int_C ds = \int_{t=a}^{t=b} \|\mathbf{v}(t)\| dt, \quad (1.3.10)$$

$$\begin{aligned} M &= \int_C \rho(x, y, z) ds \\ &= \int_{t=a}^{t=b} \rho(x(t), y(t), z(t)) \|\mathbf{v}(t)\| dt, \end{aligned} \quad (1.3.11)$$

where (1.3.11) is (1.3.6) again and (1.3.10) is the case $\rho = 1$ —the arc length (1.2.1) once more. Assume throughout that the thickness of the wire can be ignored.

Checkpoint 1.3.14 Reading the two formulas. In (1.3.10) and (1.3.11), what do a , b , t , C , dt , $\mathbf{v}$ and $\rho(x, y, z)$ represent?

a, b, t :

C, dt :

$\mathbf{v}$:

$\rho(x, y, z)$:

Checkpoint 1.3.15 The two line elements. Take two nearby points of the wire, the ones reached at the parameter values t_{k-1} and t_k . Write $\Delta \mathbf{r} = \mathbf{r}(t_k) - \mathbf{r}(t_{k-1})$ for the displacement from the first to the second, and Δs_k for the length of the piece of wire joining them. Intuitively speaking, what do ds and $d\mathbf{r}$ represent, and what happens to Δs_k and $\Delta \mathbf{r}$ in the limit that produces them?

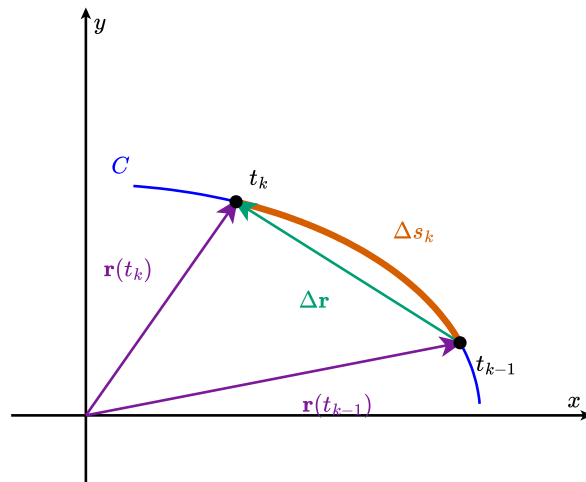


Figure 1.3.16 Two nearby points of a curve C , at the parameter values t_{k-1} and t_k . The straight arrow is the displacement $\Delta \mathbf{r} = \mathbf{r}(t_k) - \mathbf{r}(t_{k-1})$; the thick piece of curve joining the same two points has length Δs_k . The chord is straight and the arc is not, so the two differ—but less and less as the points come together.

(Say which of the two is a number and which a vector, and what each one records about the step. Then take the limit in [Figure 1.3.16](#): divide $\Delta \mathbf{r}$ by Δt_k and let the points come together; do the same for the arc, using the fact that the ratio of arc to chord tends to 1. Which substitution from earlier in the section is this?)

$d\mathbf{r}$ represents / ds represents

$\frac{\Delta \mathbf{r}}{\Delta t_k} \rightarrow$ so $\Delta \mathbf{r} \rightarrow$

$\Delta s_k \rightarrow$

Checkpoint 1.3.17 Units. If we use meters and kilograms as the units of length and mass respectively, then what will the units of $\rho(x, y, z)$ and ds be?

units of ρ : units of ds : units of ρds :

Checkpoint 1.3.18 What is being integrated? In the above applications of line integrals of scalar functions $\int_C f(x, y, z) ds$, what is the scalar function that is being integrated (the integrand) along the wire?

for the mass, $f =$ for the length, $f =$

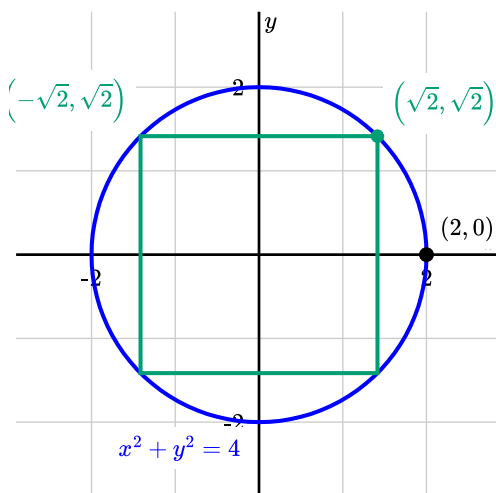


Figure 1.3.19 Two wires: the circle of radius 2 centered at the origin, and the square inscribed in it, whose vertices $(\pm\sqrt{2}, \pm\sqrt{2})$ lie on the circle.

Checkpoint 1.3.20 A mass with no parametrization. Assume that the density of the circular wire in Figure 1.3.19 is given by $\rho(x, y) = \frac{1}{x^2 + y^2 + 1}$. Without parametrizing, find its mass.

(What is $x^2 + y^2$ along the curve? Then the integrand is a constant, and what is left is the length (1.3.10) of the wire.)

along C , $\rho =$

$$M = \int_C \rho \, ds =$$

Checkpoint 1.3.21 Does the radius matter? Assume that the mass density of a circular wire of radius R centered at the origin is given by $\rho(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$. Which statement is correct?

- A. The mass of the wire is independent of its radius R .
- B. The mass of the wire increases as R increases.
- C. The mass of the wire decreases as R decreases.

(Circle your choice and justify it: what is ρ on the circle of radius R , and what does that make M ?)

on the circle, $\rho =$ so $M =$

Checkpoint 1.3.22 Which wire is heavier? Assume that the density of the square-shaped wire and of the circular wire graphed in [Figure 1.3.19](#) is given by $\rho(x, y) = x^2 + y^2$. Then, without parametrizing the curves, choose the correct option below. Which statement is correct?

- A. The circular wire is heavier.
- B. The square-shaped wire is heavier.
- C. The two wires have the same mass.

(Circle your choice and justify it. On the circle, $x^2 + y^2$ is constant, so its mass is that constant times its circumference. For the square you do not need the exact mass: the square is inscribed in the circle, so bound $x^2 + y^2$ on the square from above and multiply by its perimeter. Is the bound strict?)

$M_{\text{circle}} =$

$M_{\text{square}} <$

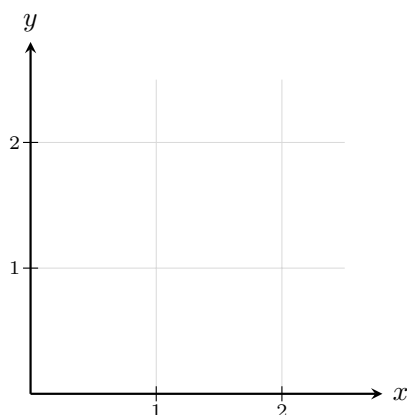
conclusion:

Checkpoint 1.3.23 A charged wire. Assume that the charge density of the square-shaped wire and of the circular wire graphed in [Figure 1.3.19](#) is given by $\rho(x, y) = |x| - |y|$. Then, without parametrizing the curves, choose the correct option(s) below. Select all that apply.

- A. The circular wire is negatively charged.
- B. The square-shaped wire is positively charged.
- C. In both cases (square and circular wires) the amount of positive charge is equal to the amount of negative charge.

(Draw the quarter of each wire lying in the first quadrant, where $\rho = x - y$, together with the line $y = x$, and mark where the density is positive and where it is negative. Now reflect in that line, that is, interchange x and y : what does

the reflection do to each shape, to lengths, and to ρ ? Conclude what the charge on that quarter is, and then on the whole wire.)



sign of ρ below and above $y = x$:

under the reflection, $\rho(y, x) =$

conclusion, and which option(s):

1.4 Vector Fields and Line Integrals

In this section we meet two ideas. A *vector field* attaches a vector—a magnitude and a direction—to every point of the plane or of space; think of the velocity of the air at each point of a room, or the gravitational pull at each point around a planet. The *line integral of a vector field* then answers a natural question: as an object moves along a curve through the field, how much does the field push it along its way? The physical name for this accumulated push is *work*.

1.4.1 Vector Fields

What is a vector field? We will study the definition through an example. Assume that we have a function $\mathbf{F}(x, y)$ on $\mathbb{R}^2$: the function $\mathbf{F}$ takes two real numbers (x, y) as its input and outputs a vector in $\mathbb{R}^2$. As an example, consider the following vector field:

$$\mathbf{F}(x, y) = \langle x, y \rangle. \quad (1.4.1)$$

To visualize this vector field, imagine that at each point (x, y) there is a vector $\langle x, y \rangle$ attached. Physically, this vector could correspond to the velocity of an atom located at the point (x, y) . See [Figure 1.4.1](#).

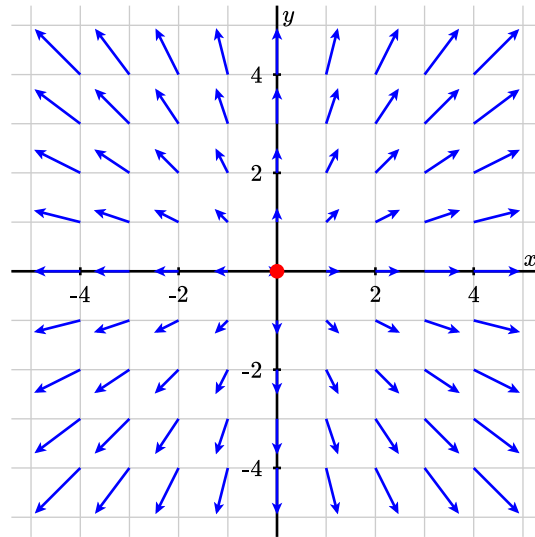


Figure 1.4.1 The vector field $\mathbf{F}(x, y) = \langle x, y \rangle$. Every arrow points directly away from the origin, and the arrows get longer as the distance from the origin grows. The arrows are drawn at 0.22 times their true length, so that they fit in the picture.

What is the length of the vector in (1.4.1)? Compute it, and use it to explain why the blue arrows in Figure 1.4.1 grow in length as we move away from the origin along a straight line. Then modify the vector field so that the length of the vectors is one at every point.

workspace

Visualized, the normalized field looks like Figure 1.4.2: the directions are unchanged, but now all arrows have the same length.

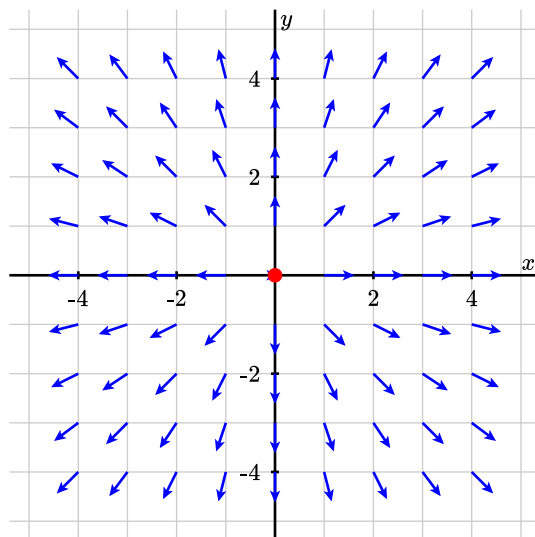


Figure 1.4.2 The normalized field $\mathbf{F}(x, y) = \frac{\langle x, y \rangle}{\sqrt{x^2 + y^2}}$: same directions as Figure 1.4.1, but every arrow has length one. The arrows are drawn with length 0.55 rather than 1, so that neighboring arrows do not overlap.

Our next example is a vector field which is a function on $\mathbb{R}^3$. Consider the following vector field:

$$\begin{aligned}\mathbf{F}(x, y, z) &= \frac{\langle -x, -y, -z \rangle}{(x^2 + y^2 + z^2)^{3/2}} \\ &= -\frac{\mathbf{r}}{\|\mathbf{r}\|^3}, \quad \mathbf{r} = \langle x, y, z \rangle.\end{aligned}\tag{1.4.2}$$

This field has the same dependence on x , y and z as the gravitational force exerted by a massive body located at the origin.¹ Every arrow points *toward* the origin. Compute $\|\mathbf{F}\|$ in terms of $\|\mathbf{r}\|$ to see how the pull weakens with distance. See Figure 1.4.3. We will meet this field again in Example 1.4.10.

workspace

¹The gravitational force exerted by a mass m_1 at the origin on a mass m_2 at the point $\mathbf{r}$ is the **gravitational force field** $\mathbf{F}_{\text{grav}} = -\frac{Gm_1m_2}{\|\mathbf{r}\|^3} \mathbf{r}$, where G is the gravitational constant and the minus sign makes the force point back toward the origin. The field $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ studied here is that force with the constant Gm_1m_2 stripped off, so the two differ only by a positive factor and point the same way at every point.

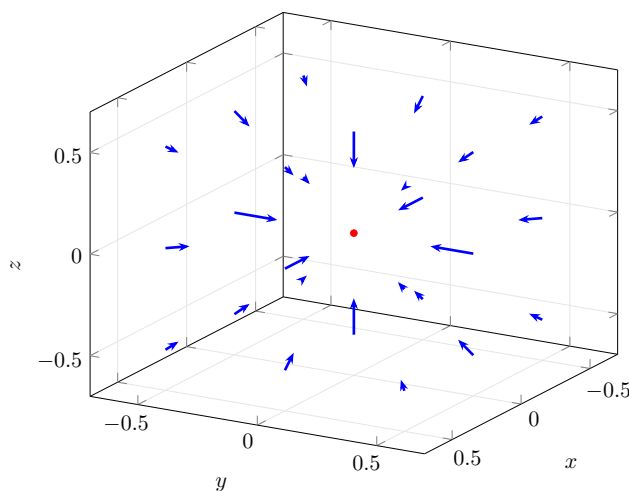


Figure 1.4.3 The field $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$, which depends on x , y and z exactly as the gravitational force does. All arrows point toward the origin and shrink with the square of the distance.

1.4.2 Work (Reminder)

Our goal is to study the line integral of a vector field over a curve C , and we will do so using an example from physics: calculating the amount of work done by a force $\mathbf{F}$, which in general can be a function of x , y , and z . Before we present the example, let us remind ourselves of the definition of work. Consider a box being pulled by a constant force $\mathbf{F}$, as shown in [Figure 1.4.4](#).

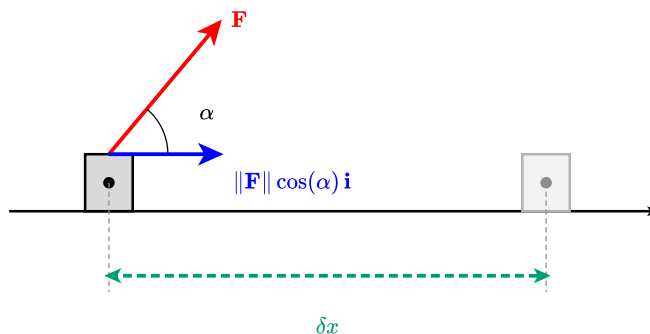


Figure 1.4.4 A box pulled by a constant force $\mathbf{F}$ applied at an angle α . The displacement δx is measured below the ground line, from the centre of mass of the box in its initial position to its centre of mass in the final position (the faint box). Only the component $\|\mathbf{F}\| \cos(\alpha) \mathbf{i}$ along the motion does work.

The amount of work done by the force is

$$\text{Work} = \underline{\hspace{10em}} \quad (1.4.3)$$

Write it first in terms of $\|\mathbf{F}\|$, α and δx , then as a dot product; the dot-product form is the one that generalizes.

1.4.3 Work of a Force Field

Let us now consider the situation where a force $\mathbf{F}(x, y)$ is acting on an object and causing it to move along a curved trajectory from an initial point to a final

point. Neither the force nor the direction of motion is constant anymore, so we cannot apply (1.4.3) directly. Instead, we reuse the strategy of Section 1.3: calculate the work on small subarcs of the trajectory, where force and direction are nearly constant, and then sum up. Figure 1.4.5 shows one such subarc.

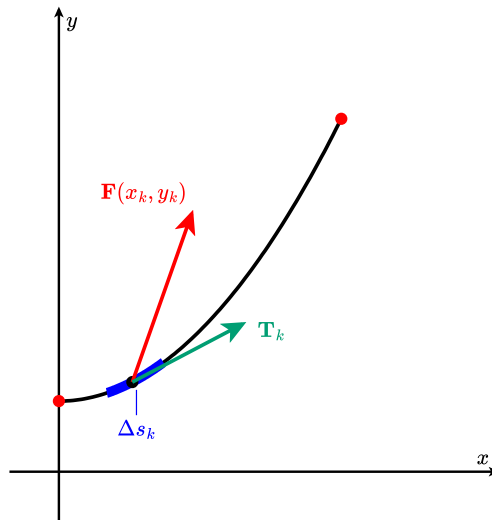


Figure 1.4.5 On the k th subarc, of length Δs_k , the direction of motion is the unit tangent vector $\mathbf{T}_k$ (green) and the force is approximately $\mathbf{F}(x_k, y_k)$ (red). Only the tangential component of the force does work.

Build the work integral from the subarcs, down to its form in terms of a parametrization $\mathbf{r}(t)$, $a \leq t \leq b$.

workspace

The Line Integral of a Vector Field.

The work done by $\mathbf{F}$ along C is approximated by

$$W \approx \sum_{k=1}^n \mathbf{F}(x_k, y_k) \cdot \mathbf{T}_k \Delta s_k, \quad (1.4.4)$$

and exactly by

$$W = \int_C \mathbf{F} \cdot \mathbf{T} ds, \quad (1.4.5)$$

where $\mathbf{T}$ is the unit tangent vector of the curve. Since

$$\mathbf{T} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\frac{d\mathbf{r}}{dt}}{\frac{ds}{dt}} = \frac{d\mathbf{r}}{ds}, \quad (1.4.6)$$

the combination $\mathbf{T} ds$ is simply $d\mathbf{r}$, and

$$W = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_C \mathbf{F} \cdot d\mathbf{r}. \quad (1.4.7)$$

In practice, once the curve is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{v}(t) dt : \quad (1.4.8)$$

evaluate the field along the curve, dot it with the velocity, and integrate over time.

The applet below carries every ingredient of (1.4.8) at once: a curve C parametrized by $\mathbf{r}(t)$, the velocity $\mathbf{v}(t) = d\mathbf{r}/dt$ tangent to it, and a force field $\mathbf{F}$ evaluated at the moving point. Work out what the integrand $\mathbf{F} \cdot \mathbf{v}$ is doing before you compute anything.

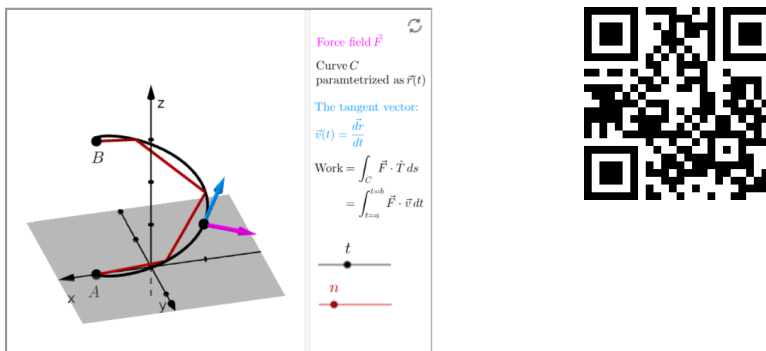


Figure 1.4.6 A point climbs the curve C while the force field $\mathbf{F}$ and the velocity $\mathbf{v}$ are drawn at its current position. The work is the accumulated $\mathbf{F} \cdot \mathbf{v}$, so the angle between the two arrows is what decides its sign.

Work was our motivation, but any vector field $\mathbf{F}$ can be integrated over a curve C in this fashion; the number it produces is also called the **circulation** of $\mathbf{F}$ along C , a name that becomes the usual one when C is a closed curve. Figure 1.4.7 shows the accumulation of $\mathbf{F} \cdot \mathbf{T} ds$ along a trajectory.



Figure 1.4.7 As the object moves through the field $\mathbf{F} = \langle x, y \rangle$ along the trajectory, the work $W = \int_C \mathbf{F} \cdot d\mathbf{r}$ accumulates moment by moment from the tangential component of the force.

There is a third, frequently used way to write the same integral. Assume the vector field has components $\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z) \rangle$. Since

$$d\mathbf{r} = \underline{\hspace{2cm}},$$

expanding the dot product gives the **differential form** of the line integral:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \underline{\hspace{4cm}} \quad (1.4.9)$$

All three notations—(1.4.5), $\int_C \mathbf{F} \cdot d\mathbf{r}$, and the differential form—mean exactly the same thing.

Example 1.4.8 Work along a twisted cubic. Calculate the work done by the force

$$\mathbf{F}(x, y, z) = \langle y - x^2, z - y^2, x - z^2 \rangle$$

over the curve C :

$$\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$$

from $(0, 0, 0)$ to $(1, 1, 1)$.

Solution.

workspace

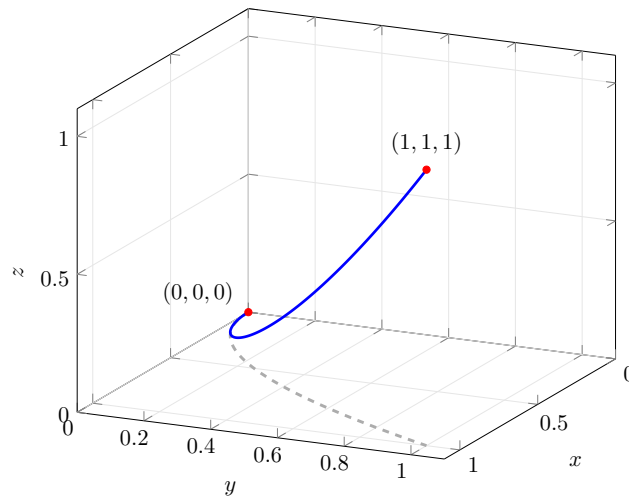


Figure 1.4.9 The twisted cubic $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ from the origin to $(1, 1, 1)$. The dashed gray curve is its shadow on the plane $z = 0$.

□

Example 1.4.10 Work done by an inverse-square field along a helix.
Calculate the line integral

$$\int_C \frac{-x}{(x^2 + y^2 + z^2)^{3/2}} dx + \frac{-y}{(x^2 + y^2 + z^2)^{3/2}} dy$$

$$+ \frac{-z}{(x^2 + y^2 + z^2)^{3/2}} dz$$

over the curve $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$.

Solution.

workspace

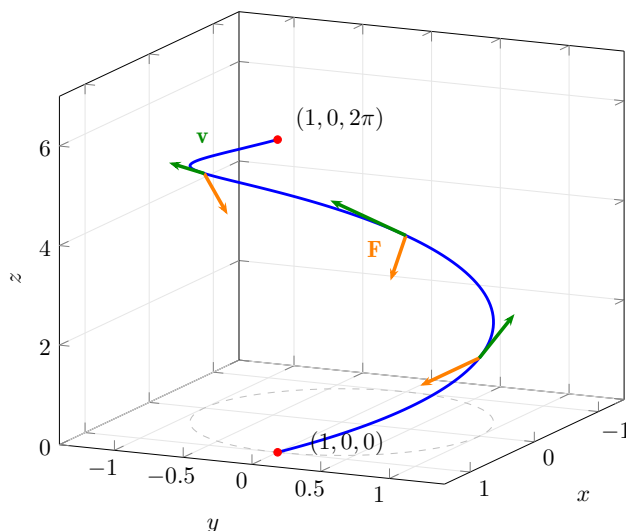


Figure 1.4.11 The helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$, climbing through the field $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ (orange arrows, direction only). At the same three points the velocity $\mathbf{v}$ is drawn in green; compare the angle between the two arrows with the sign of your answer.

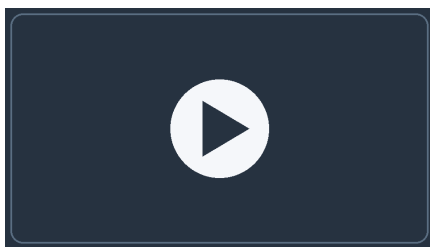


Figure 1.4.12 A particle climbs the helix through the field while the work $W(t)$ accumulates. Watch its sign, and check the value it reaches at the top against your answer.

□

Example 1.4.13 The same endpoints, two paths, two answers. Calculate the work done by the force

$$\mathbf{F}(x, y) = \langle y, 2x \rangle$$

in moving a particle from $O(0, 0)$ to $P(1, 1)$ along each of the following two paths:

- C_1 , the straight segment from O to P , parametrized by $\mathbf{r}_1(t) = \langle t, t \rangle$, $0 \leq t \leq 1$;
- C_2 , the arc of the parabola $y = x^2$ from O to P , parametrized by $\mathbf{r}_2(t) = \langle t, t^2 \rangle$, $0 \leq t \leq 1$.

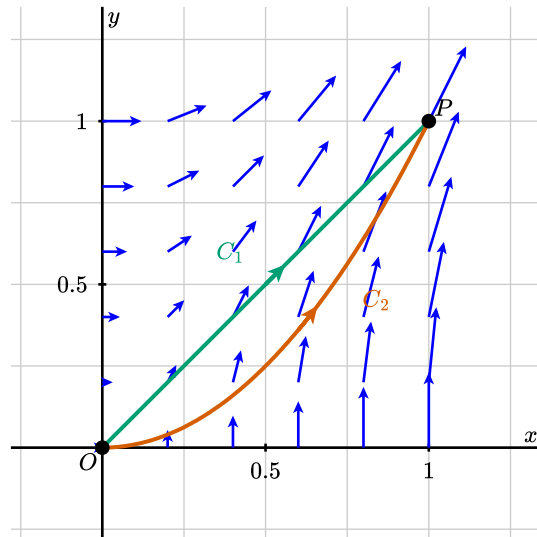


Figure 1.4.14 The two paths of [Example 1.4.13](#) drawn in the field $\mathbf{F} = \langle y, 2x \rangle$: the segment C_1 along $y = x$ and the arc C_2 along $y = x^2$, both running from $O(0,0)$ to $P(1,1)$.

Solution.

workspace

□

1.4.4 Concept Check

Checkpoint 1.4.15 Positive, negative, or zero? In Figure 1.4.16, determine whether the work done by the force field along the two paths C_1 and C_2 is positive, negative, or zero.

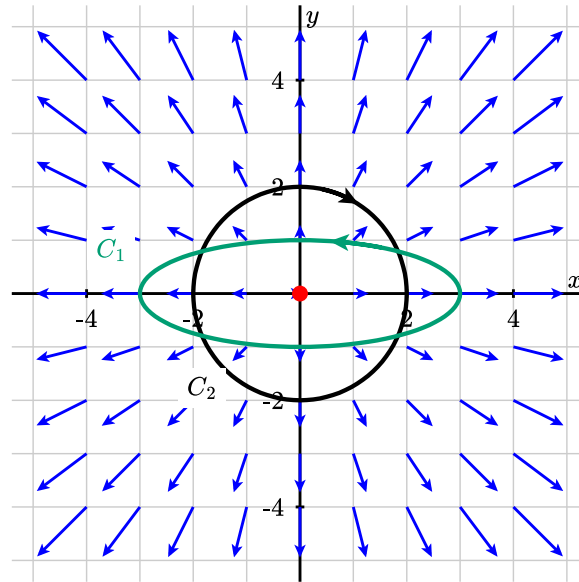


Figure 1.4.16 Two closed paths in the radial field (1.4.1): the ellipse C_1 and the circle C_2 . Notice the orientation of each curve.

workspace

Checkpoint 1.4.17 A force at a fixed angle to the path. An object moves along a circle of radius R centred at O and is acted upon by a force $\mathbf{F}$ of constant magnitude $F = \|\mathbf{F}\|$. At every moment the force makes a 30° angle with the unit tangent vector $\mathbf{T}$, as shown in Figure 1.4.18. Calculate the work done by $\mathbf{F}$ as the object moves along the half circle C from A to B .

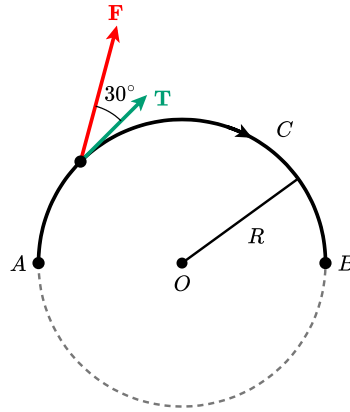


Figure 1.4.18 The object travels over the top of the circle from A to B . At every point the force $\mathbf{F}$ (red) makes a 30° angle with the unit tangent vector $\mathbf{T}$ (green), the direction of travel.

workspace

Checkpoint 1.4.19 Pulling a pendulum aside. A simple pendulum consists of a small bob of mass m hanging from a cord of length ℓ whose mass is negligible. A horizontal force $\mathbf{F} = F\mathbf{i}$ moves the bob so slowly that its acceleration is essentially zero; its magnitude $F = \|\mathbf{F}\|$ must therefore change with the angle θ that the cord makes with the vertical. Two other forces act on the bob: gravity, $\mathbf{F}_G = m\mathbf{g}$, and the force $\mathbf{F}_T$ that the cord exerts on it. See [Figure 1.4.20](#).

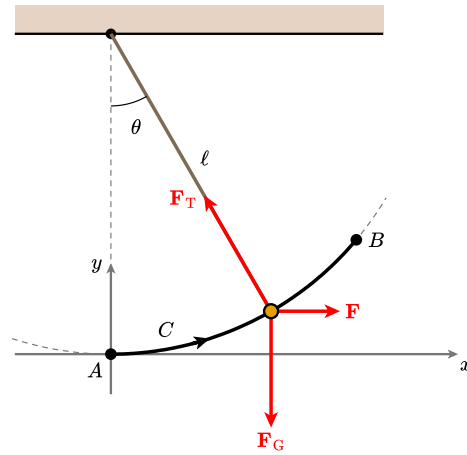


Figure 1.4.20 The bob moves along the arc C from its lowest point A to the point B where the cord makes the angle θ_0 with the vertical. Three forces act on it: the horizontal pull $\mathbf{F}$, gravity $\mathbf{F}_G$, and the force $\mathbf{F}_T$ of the cord.

- (a) Calculate the work done by $\mathbf{F}$ in moving the pendulum from $\theta = 0$ to $\theta = \theta_0$.

workspace

- (b) Calculate the work done by the gravitational force $\mathbf{F}_G$ and the work done by the force $\mathbf{F}_T$ of the cord over the same motion.

workspace

1.5 Path Independence, Conservative Fields, and Potential Functions

Before we begin discussing the actual topic of today's lecture, we will need to study a particular type of a vector field known as a *gradient field*.

1.5.1 A Brief Reminder (Gradient Fields)

Definition 1.5.1 Gradient Field. The **gradient field** of a differentiable function $f(x, y, z)$ is the field of gradient vectors

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}. \quad (1.5.1)$$

◇

At each point (x, y, z) , the gradient field gives a vector pointing in the direction of greatest increase of f . Let us emphasize that even though the function f is a scalar-valued function, its gradient is a vector-valued one by definition. As an example, see [Figure 1.5.2](#).

$$z = f(x, y) = x^2 + y^2$$

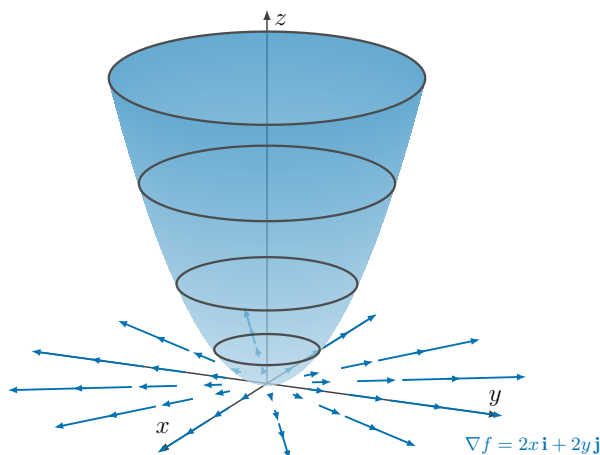


Figure 1.5.2 The surface $z = f(x, y) = x^2 + y^2$ together with its gradient field $\nabla f = 2x\mathbf{i} + 2y\mathbf{j}$, drawn in the xy -plane. The gradient vectors point radially away from the origin, the direction of steepest ascent, and grow longer as we move away from the origin.

In Figure 1.5.2, you see the scalar-valued function $z = f(x, y) = x^2 + y^2$, which is a function on $\mathbb{R}^2$. Compute its gradient according to (1.5.1), and compare with the arrows in the figure.

$\nabla f =$

The gradient field is visualized in the xy -plane and the value of the function $f(x, y) = x^2 + y^2$ at each point (x, y) is plotted on the z -axis. Intuitively, the gradient of a function $f(x, y)$ at a point (x_0, y_0) is a vector pointing in the direction of the steepest slope.

1.5.2 Path Independence and Some Basic Definitions

Previously, in Example 1.4.13, we saw that the line integral of a vector field depends on the path in general. Then, the natural question to ask is:

What are the conditions under which the line integral of a vector field $\mathbf{F}$ from point A to point B is independent of the curve C taken to go from A to B ?

Answering this question requires a few new ideas, so we introduce them first.

Remark 1.5.3 Reminder. A **smooth curve** has no sharp corners or cusps; when the tangent vector turns, it does so continuously.

Definition 1.5.4 Open, Connected, and Simply Connected Regions; Conservative Fields.

- A region D is **open** if for every point P in D there is a disk with center P that lies entirely in D , which means that D doesn't contain any of its boundary points.

- Consider an open region D in space and a vector field $\mathbf{F}$. If the result of the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ from A to B is the same for all possible paths inside D , then the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is said to be **path-independent** inside D and the field $\mathbf{F}$ is said to be **conservative** on D .
- If a vector field $\mathbf{F}$ can be written as $\mathbf{F} = \nabla f$, where f is a scalar-valued function, then f is a **potential function**.
- **Piecewise smooth** curves are curves that are made up of finitely many smooth pieces.
- An open region D is said to be **connected** if any two points in D can be joined by a smooth curve that lies in the region.
- D is **simply connected**, which means that every loop in D can be contracted to a point in D without ever leaving D . (See Figure 1.5.5.)

◇

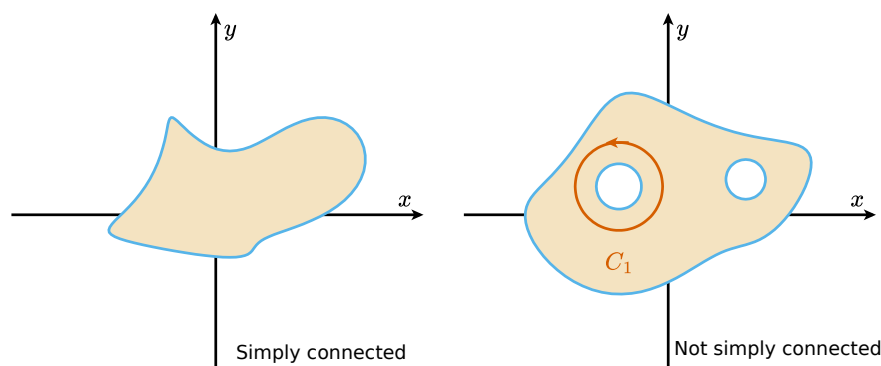


Figure 1.5.5 On the left, a simply connected region: every loop can be contracted to a point without leaving the region. On the right, a region that is not simply connected: the loop C_1 encircles a hole and cannot be contracted to a point while staying inside the region.

1.5.3 Line Integrals in Conservative Fields

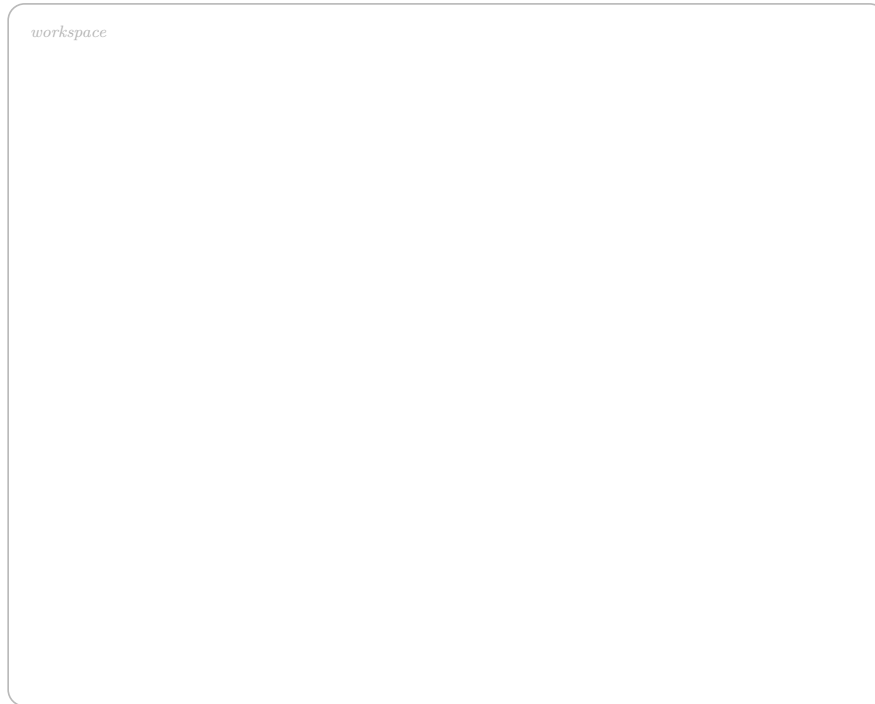
Next, we will study the line integrals of conservative fields, in the sense of Definition 1.5.4. As you will recognize, if you think of the gradient of a vector field as its derivative, then the fundamental theorem of line integrals makes a similar statement for vector-valued functions as the one made for definite integrals of scalar-valued functions, i.e., the one that you know as FTC II.¹

Theorem 1.5.6 Fundamental Theorem of Line Integrals. *Let C be a smooth curve joining the point A to the point B in the plane or in space and parametrized by $\mathbf{r}(t)$. Let f be a differentiable function with a continuous gradient vector $\mathbf{F} = \nabla f$ on a domain D containing C . Then*

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A). \quad (1.5.2)$$

¹The Fundamental Theorem of Calculus, Part II: if f is continuous on $[a, b]$ and F is any antiderivative of f on that interval, then $\int_a^b f(x) dx = F(b) - F(a)$. It evaluates a definite integral from the values of an antiderivative at the two endpoints, and Theorem 1.5.6 does the same for a line integral, using the values of a potential function at the two endpoints of the curve.

Proof.



In [Figure 1.5.7](#), the region D is drawn as a translucent solid and the curve C is in red. According to the fundamental theorem, the line integral of a conservative vector field $\mathbf{F}$ can be easily calculated by calculating the values of its corresponding potential function f ([Definition 1.5.4](#)) at the beginning (point A) and end (point B) of the curve C . ■

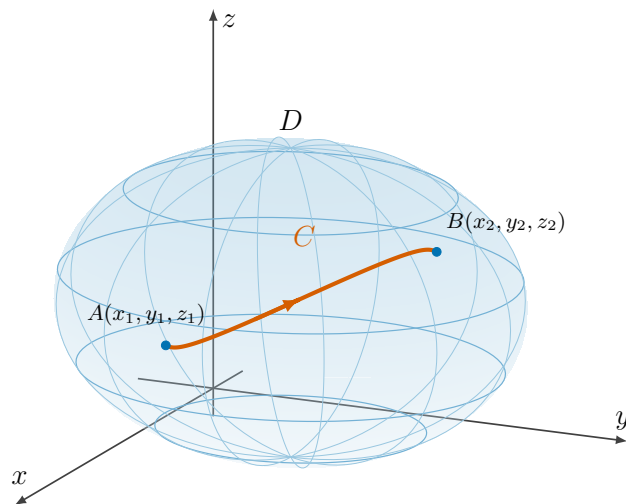


Figure 1.5.7 A smooth curve C joining $A(x_1, y_1, z_1)$ to $B(x_2, y_2, z_2)$ inside an open region D in space, drawn as a translucent solid containing C . For a conservative field $\mathbf{F} = \nabla f$, the line integral along C equals $f(B) - f(A)$, no matter how C winds from A to B .

The central idea of the theorem—that only the endpoints matter—is animated in [Figure 1.5.8](#).

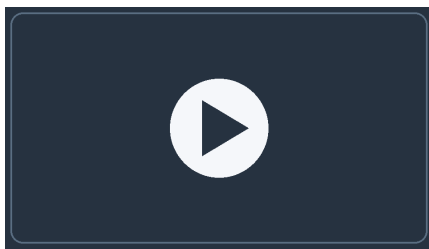


Figure 1.5.8 Animation of the fundamental theorem of line integrals for the field $\mathbf{F} = \nabla f$ with $f(x, y, z) = 1/\sqrt{x^2 + y^2 + z^2}$. Two different curves join $(1, 0, 0)$ to $(1, 0, 2\pi)$: the helix C_1 and the straight segment C_2 . The accumulated value of each line integral is tracked as the paths are traced, and both integrals arrive at the same value $f(B) - f(A)$.

In the next example, we will see how we can use [Theorem 1.5.6](#).

Example 1.5.9 Example I. Consider the scalar-valued function $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$.

- A. Calculate the gradient vector field $\mathbf{F}(x, y, z) = \nabla f$.
- B. Evaluate the line integral of the vector field $\mathbf{F}(x, y, z) = \nabla f$ over the curve C_1 parametrized as $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$.
- C. What is the result of $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$ where C_2 is the line connecting the two points?

Solution.

workspace

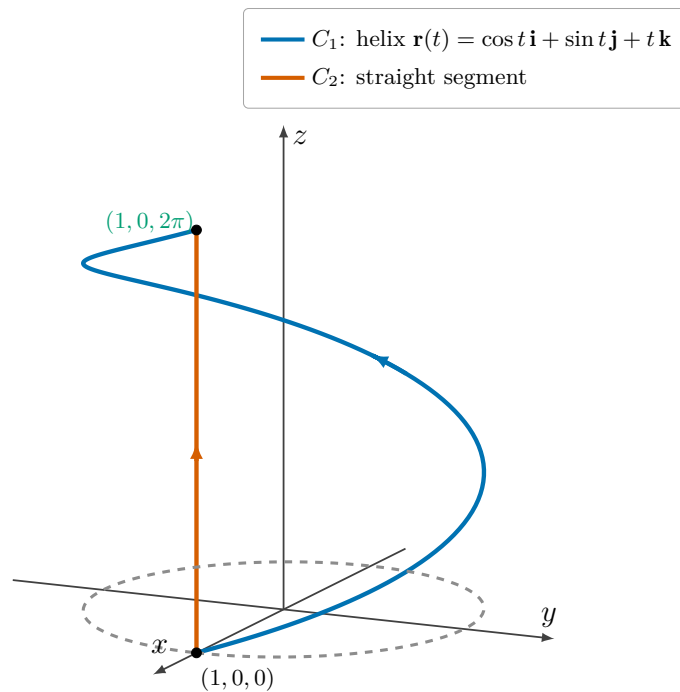


Figure 1.5.10 The two paths of [Example 1.5.9](#) joining $(1, 0, 0)$ to $(1, 0, 2\pi)$: the helix C_1 , parametrized by $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$, and the straight segment C_2 .

1.5.4 Finding a Potential Function

Example 1.5.11 Example II. Consider the vector field $\mathbf{F} = (2xy + 3)\mathbf{i} + (x^2 - 4y)\mathbf{j}$ in the plane.

- Find a function f such that $\mathbf{F} = \nabla f$.
- Use the solution of part (A) to calculate the line integral of $\mathbf{F}$ along any curve from $(0, 0)$ to $(1, 2)$.

Solution.

workspace

□

Example 1.5.12 Example III. Consider the vector field $\mathbf{F} = yz\mathbf{i} + xz\mathbf{j} + (xy + 2z)\mathbf{k}$.

- A. Find a function f such that $\mathbf{F} = \nabla f$.
- B. Use the solution of part (A) to calculate the line integral from $(1, 0, -2)$ to $(4, 6, 3)$.

Solution.

workspace

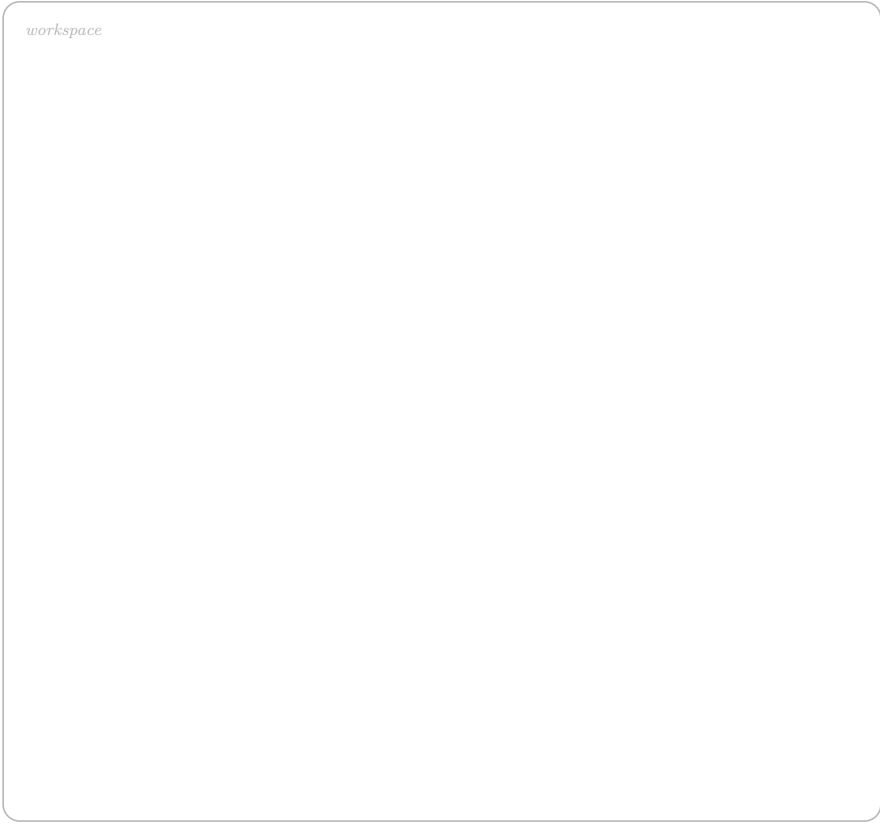
□

Example 1.5.13 Example IV. In [Example 1.4.10](#) we computed the work done by the inverse-square field

$$\mathbf{F} = \frac{\langle -x, -y, -z \rangle}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\mathbf{r}}{\|\mathbf{r}\|^3}$$

along the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$, by parametrizing the curve. Do it again with [Theorem 1.5.6](#).

Solution.



workspace

□

1.5.5 Conservative Fields are Gradient Fields

Theorem 1.5.14 **Conservative Fields are Gradient Fields.** *Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ be a vector field whose components are continuous throughout an open connected region D in space ([Definition 1.5.4](#)). Then $\mathbf{F}$ is conservative if and only if $\mathbf{F}$ is a gradient field ∇f for a differentiable function f .*

Proof.

workspace

Path independence means that all of the paths in [Figure 1.5.15](#) give the same value of the line integral: if $\mathbf{F}$ is conservative, then ■

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_3} \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A). \quad (1.5.3)$$

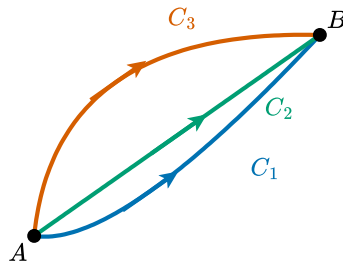


Figure 1.5.15 Three different paths C_1 , C_2 , and C_3 joining A to B . For a conservative field, all three line integrals have the same value $f(B) - f(A)$.

1.5.6 The Loop Property of Conservative Fields

Theorem 1.5.16 Loop Property of Conservative Fields. *The following statements are equivalent.*

1. $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ around every loop (that is, closed curve C) in D .
2. The field $\mathbf{F}$ is conservative on D .

Proof.

workspace



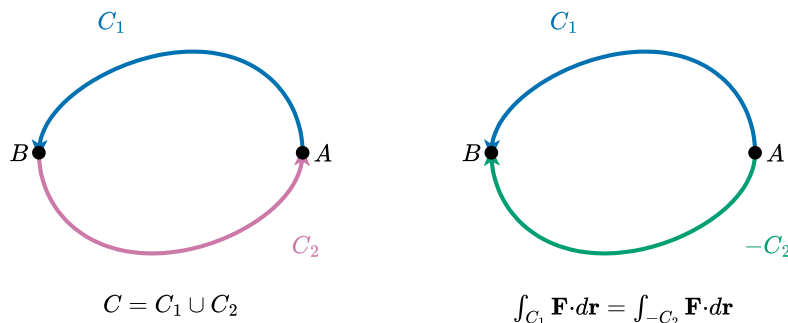


Figure 1.5.17 On the left, a loop C is split into two pieces: C_1 from A to B and C_2 from B back to A , so $C = C_1 \cup C_2$. On the right, the orientation of C_2 is reversed to produce $-C_2$; now C_1 and $-C_2$ both connect A to B , and path independence forces the two integrals to agree.

We can now collect everything this section has proved. The content of [Theorem 1.5.14](#), combined with [Theorem 1.5.6](#), the loop property in [Theorem 1.5.16](#), and the definition of path independence in [Definition 1.5.4](#), can be organized into the following square of equivalent statements. Throughout, $\mathbf{F}$ is a continuous field with continuous partial derivatives on an open, connected, simply connected region D .

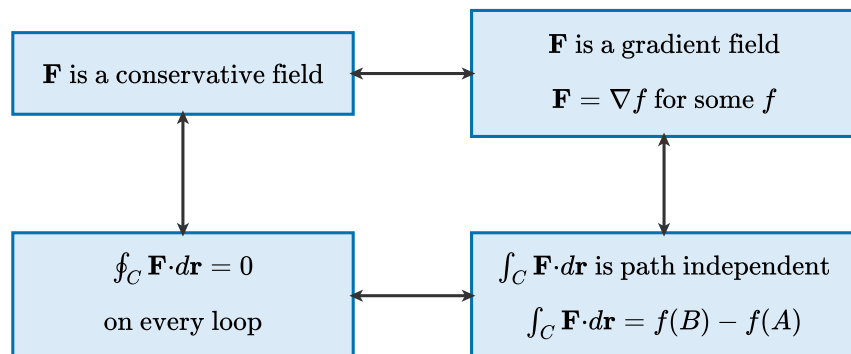


Figure 1.5.18 The four statements are equivalent. Each double arrow joins the middle of one box to the middle of the next, and may be read in either direction.

Read across the top: a conservative field is exactly a gradient field, and the potential function f is what makes that concrete. Read across the bottom: path independence and the loop property are two ways of saying the same thing, and the Fundamental Theorem turns either one into a number, $f(B) - f(A)$. Read down either side: each statement on the top implies, and is implied by, the one beneath it. In practice the four are used in whichever direction is convenient—find a potential function to evaluate a line integral, or observe that a field is a gradient field to conclude at once that its integral around every loop vanishes.

1.5.7 Concept Check

The three questions below all use the applets that go with them. In each one, drag the slider to send the point along the curve and watch the field vector and the tangent vector move with it; the sign of their dot product is what the integral is accumulating. Every applet can be rotated by dragging the background and zoomed by scrolling.

Checkpoint 1.5.19 Work Along the Helix, Two Ways. For the field and path of [Figure 1.5.20](#):

1. Use the parametrized path to compute the work done by the force field from point A to point B .
2. Is it possible to use the fundamental theorem of line integrals ([1.5.2](#)) to compute that work? If so, use it to verify your answer to the previous part. Justify your answer fully.

workspace

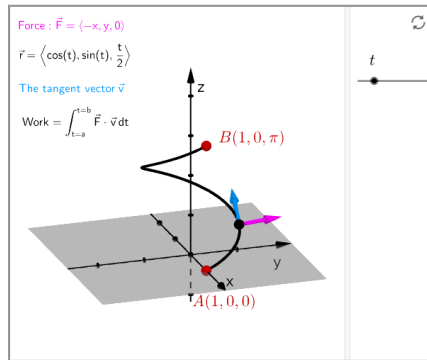


Figure 1.5.20 A point climbing the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t/2 \rangle$ from A to B , through the force field $\mathbf{F} = \langle -x, y, 0 \rangle$. The two arrows are $\mathbf{F}$ at the moving point and the velocity $\mathbf{v}$.

Checkpoint 1.5.21 A Line Integral Along a Segment. Compute the line integral of the vector field along the line segment shown in [Figure 1.5.22](#).

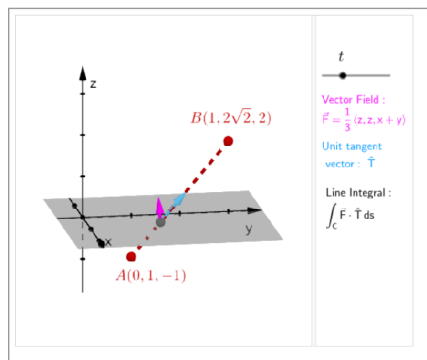
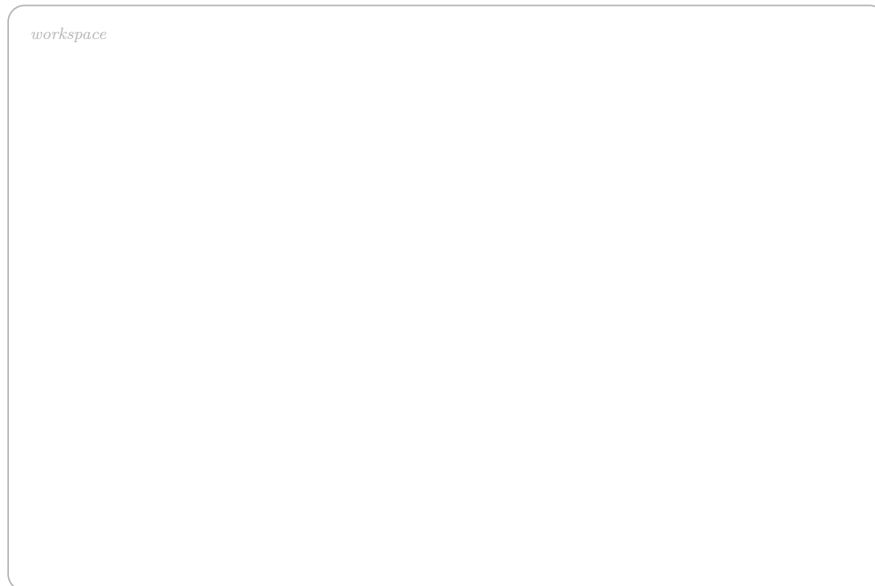


Figure 1.5.22 The straight segment from $A = (0, 1, -1)$ to $B = (1, 2\sqrt{2}, 2)$ in the field $\mathbf{F} = \frac{1}{3} \langle z, z, x + y \rangle$, with $\mathbf{F}$ and the unit tangent $\hat{\mathbf{T}}$ drawn at the moving point.

Checkpoint 1.5.23 The Same Endpoints, a Wilder Path. Compute the line integral of the vector field from point A to point B as shown in Figure 1.5.24, where the curve is parametrized as

$$\mathbf{r}(t) = \left\langle \frac{\ln(t)}{\ln(2)}, t^{3/2}, t \cos(\pi t) \right\rangle.$$

workspace

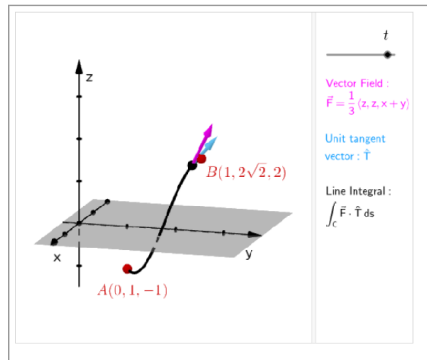


Figure 1.5.24 The curve $\mathbf{r}(t) = \left\langle \frac{\ln t}{\ln 2}, t^{3/2}, t \cos(\pi t) \right\rangle$ from $A = (0, 1, -1)$ to $B = (1, 2\sqrt{2}, 2)$ in the field $\mathbf{F} = \frac{1}{3} \langle z, z, x + y \rangle$.